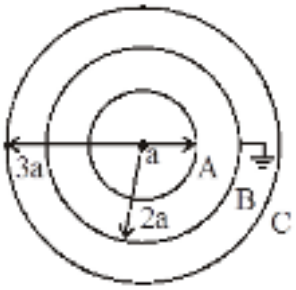
PHYSICS

1) Figure shows a system of three concentric metal shells A, B and C with radii a, 2a and 3a respectively. Shell B is earthed and shell C is given a charge Q. Now if shell C is connected to shell A, then the final charge on the shell B, is equal to :-



(1) $-\frac{4Q}{13}$

(2) $-\frac{8Q}{11}$

(3) $-\frac{5Q}{3}$

(4) $-\frac{3Q}{7}$

2) Potential energy of a system comprising of point charges is U_1.When a charge q is added in the system without disturbing other charges, the potential energy becomes U_2.The potential of the point where the charge q is placed in the system is

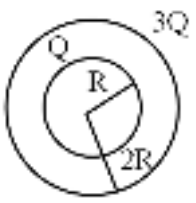
(1) $\frac{U_2-U_1}{q}$

(2) $\frac{U_1-U_2}{q}$

(3) $\frac{U_1+U_2}{2q}$

(4) $\frac{U_2-U_1}{2q}$

3) Magnitude of electric field at distance 'x' outside and inside from the surface of outer sphere is same then the value of x is –



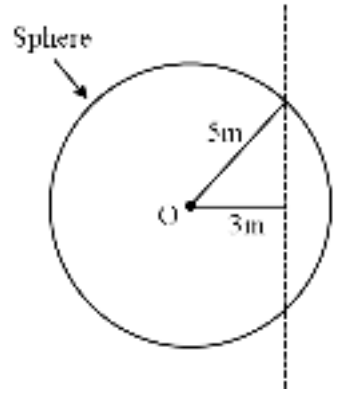
(1) R/3

(2) 2R/3

(3) $\frac{4}{3}R$

(4) $\frac{5}{3}R$

4) A charged wire of density 8.85 μC/m is passed through a sphere of radius 5m at a distance 3m from centre. as shown in fig then flux passing through the sphere will be :-



(1) $8 \times 10^5 \frac{N-m^2}{C}$

(2) $8 \times 10^6 \frac{N-m^2}{C}$

(3) $10 \times 10^3 \frac{N-m^2}{C}$

(4) None

5) At any point (x,0,0) the electric potential V is $\left(\frac{1000}{x} + \frac{1500}{x^2} + \frac{500}{x^3}\right)$ volt, then electric field at x = 1 m -

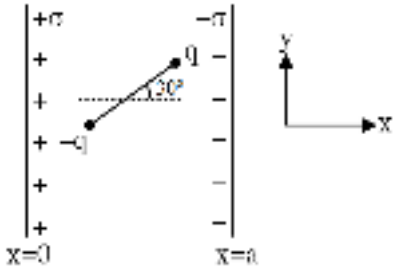
(1) $5500\left(\hat{j}+\hat{k}\right)$ V/m

(2) $5500\hat{i}$ V/m

(3) $\frac{5500}{\sqrt{2}}\left(\hat{j}+\hat{k}\right)$ V/m

(4) $\frac{5500}{\sqrt{2}}\left(\hat{i}+\hat{k}\right)$ V/m

6) An electric dipole has charges –q and q separated by d, kept at an angle 30° with x-axis between two large plane sheet of charge as shown in figure. The incorrect option is



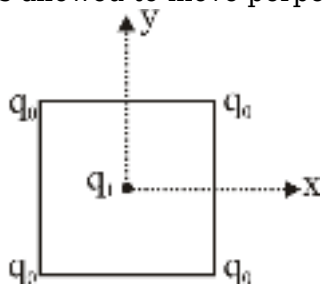
(1) Net force on dipole is zero

(2) Torque acting on dipole is $\frac{\sigma qd}{2\epsilon_0}$

(3) Potential energy of dipole is $\frac{-\sigma qd}{2\epsilon_0}\sqrt{3}$

(4) Potential energy of dipole is $\frac{-\sigma qd}{2\epsilon_0}$

7) Four equal charges q_0 each, are kept fixed on the vertices of a square of side 'a'. Charge q_1 is kept at the center of the square as shown in the figure. Then which statement is correct. (here charge q_1 is allowed to move perpendicular to XY plane only i.e. along z-axis)



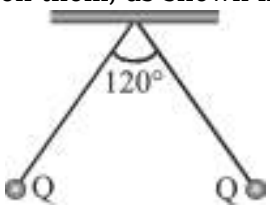
(1) charge q_1 is always in stable equilibrium.

(2) charge q_1 is always in unstable equilibrium.

(3) charge q_1 is always in stable equilibrium only if q_1 and q_0 have different (opposite) signs.

(4) charge q_1 is always in stable equilibrium only if q_1 and q_0 have same signs.

8) Two small spherical balls each carrying a charge Q = 10 μC are suspended by two insulating threads of equal lengths $\sqrt{3}$m each, from a point fixed in the ceiling. It is found that in equilibrium threads are separated by an angle 120° between them, as shown in the figure. What is the tension in the threads (Given : $\frac{1}{(4\pi\varepsilon_0)} = 9 \times 10^9$ Nm^2/C^2)



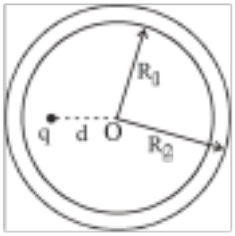
(1) $\left(\frac{0.2}{\sqrt{3}}\right)$ N

(2) 1.8 N

(3) $\left(\frac{0.2}{\sqrt{5}}\right)$ N

(4) None of the above

9) A hollow, conducting spherical shell of inner radius R_1 and outer radius R_2 encloses a charge q inside, which is located at a distance d (< R_1) from the centre of the spheres. The potential at the

centre 'O' of the shell is 

(1) Zero

(2) $\frac{1}{4\pi\varepsilon_0 d}$

(3) $\frac{1}{4\pi\varepsilon_0}\left(\frac{q}{d}-\frac{q}{R_1}\right)$

(4) $\frac{1}{4\pi\varepsilon_0}\left(\frac{q}{d}-\frac{q}{R_1}+\frac{q}{R_2}\right)$

10) Two point charge of 100 μC and –4 μC are positioned at points ($-2\sqrt{3}$, $3\sqrt{3}$, –4) and ($4\sqrt{3}$, $-5\sqrt{3}$, 6) respectively of a Cartesian coordinate system. Find the force vector on the –4 μC charge? All the coordinates are in meters.

(1) $9\times10^{-4}\left(3\sqrt{3}\hat{i}-4\sqrt{3}\hat{j}+5\hat{k}\right)$

(2) $9\times10^{-4}\left(-3\sqrt{3}\hat{i}+4\sqrt{3}\hat{j}-5\hat{k}\right)$

(3) $2.25\times10^{-4}\left(-3\sqrt{3}\hat{i}+4\sqrt{3}\hat{j}-5\hat{k}\right)$

(4) $2.25\times10^{-4}\left(3\sqrt{3}\hat{i}-4\sqrt{3}\hat{j}+5\hat{k}\right)$

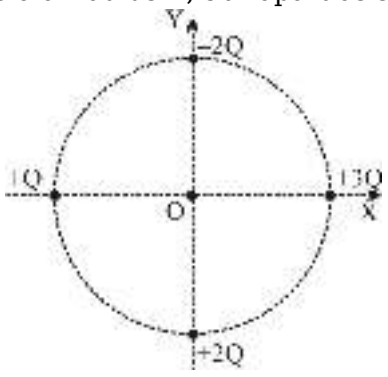
11) A sphere of radius R carries charge density proportional to the square of the distance from the center: $\rho = Ar^2$, where A is a positive constant. At a distance of R/2 from the center, the magnitude of the electric field is :-

(1) $A/4\pi\varepsilon_0$

(2) $AR^3/40\varepsilon_0$

(3) $AR^3/24\varepsilon_0$

(4) $AR^3/5\varepsilon_0$

12) Four charges are placed on the circumference of a circle of radius R, 90° apart as shown in the fig. The electric field strength at the centre of the circle is 

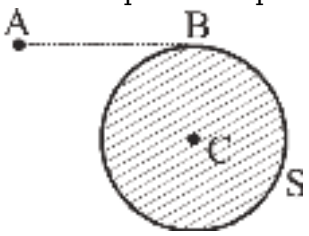
(1) $\frac{1}{4\pi\varepsilon_0}\frac{2\sqrt{5}Q}{R^2}$, making angle $\tan^{-1}2$ with the – ve axis.

(2) $\frac{1}{4\pi\varepsilon_0}\frac{2\sqrt{5}Q}{R^2}$, making angle $\tan^{-1}2$ with the + ve axis.

(3) $\frac{1}{4\pi\varepsilon_0}\frac{4\sqrt{2}Q}{R^2}$, making angle $\tan^{-1}\left(\frac{1}{2}\right)$ with the – ve axis.

(4) $\frac{1}{4\pi\varepsilon_0}\frac{4\sqrt{2}Q}{R^2}$, making angle $\tan^{-1}\left(\frac{1}{2}\right)$ with the + ve axis.

13) S is a solid neutral conducting sphere. A point charge q = 1×10^{-6} C is placed at point A. C is the centre of sphere and AB is a tangent, BC = 3 m and AB = 4 m.



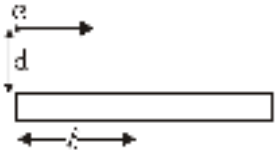
(1) The electric potential at B due to induced charge on the sphere is 1.2 kV.

(2) The electric potential at B due to induced charge on the sphere is –1.2kV.

(3) The electric potential at B due to induced charge on the sphere is –0.45kV.

(4) The electric potential at B due to induced charge on the sphere is 0.45kV.

14) An electron of mass m is projected from a distance 'd' with initial velocity u parallel to a uniformly charged flat conducting plate as shown. It strikes the plate after travelling a distance along the direction of projection, the surface charge density of conducting plate is equal to :



(1) $\frac{2d\varepsilon_0 mu^2}{e\ell^2}$

(2) $\frac{2d\varepsilon_0 mu^2}{e\ell}$

(3) $\frac{d\varepsilon_0 mu^2}{e\ell}$

(4) $\frac{d\varepsilon_0 mu}{e\ell}$

15) Let there be a spherically symmetric charge distribution with charge density varying as $\rho(r) = \rho^0\left(\frac{5}{4}-\frac{r}{R}\right)$ upto r = R, and ρ(r) = 0 for r > R, where r is the distance from the origin. The electric potential at a distance R from the origin is given by :

(1) $\frac{\rho_0 r}{3\varepsilon_0}\left(\frac{5}{4}-\frac{r}{R}\right)$

(2) $\frac{4\pi\rho_0 r}{3\varepsilon_0}\left(\frac{5}{3}-\frac{r}{R}\right)$

(3) $\frac{\rho_0 R^2}{6\varepsilon_0}$

(4) $\frac{\rho_0 R^2}{3\varepsilon_0}$

16) Two charges –9μC and +9μC are placed at the points P (1, 0, 4) m and Q (2, –1, 5) m located in the electric field $\vec{E} = 0.20\hat{i}\,\frac{V}{cm}$. Calculate torque acting on the dipole.

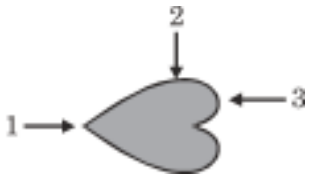
(1) 25×10^{-7} N - m

(2) 25×10^{-5} N - m

(3) 15×10^{-7} N - m

(4) 15×10^{-5} N - m

17) A heart shaped conductor shown below carries net charge Q. Which of the statement about the electric field E and the surface charge density σ below is correct?



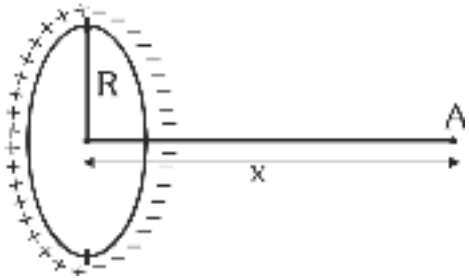
(1) E strongest and σ smallest at position-1

(2) E strongest and σ highest at position-1

(3) E weakest and σ highest at position-2

(4) E strongest and σ highest at position-3

18) Half of the non–conducting ring has (+Q) charge and half has (–Q) charge. Find potential at point A which is on the line passing through centre perpendicular to plane of ring.



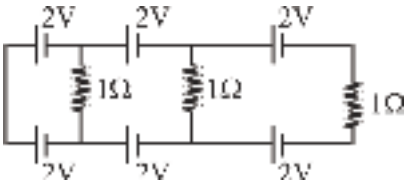
(1) $\frac{KQ}{\sqrt{R^2+x^2}}$

(2) $\frac{2KQ}{\sqrt{R^2+x^2}}$

(3) 0

(4) None of these

19) In the below circuit the current in each resistance is :



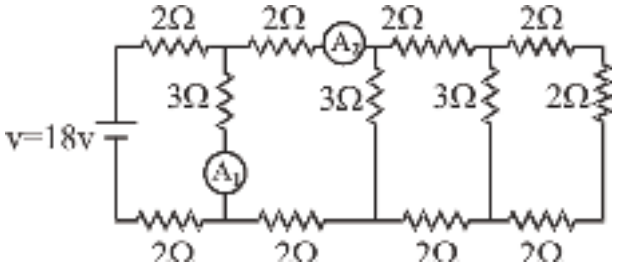
(1) 0.5 A

(2) 0 A

(3) 1 A

(4) 0.25 A

20) Reading of A_1 and A_2 are respectively



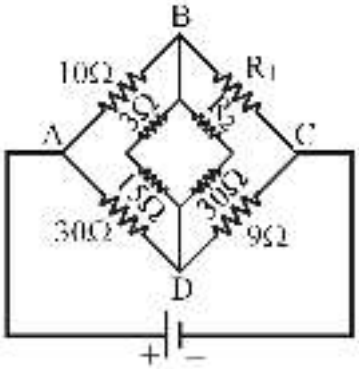
(1) 2A , 1A

(2) 1A , 2A

(3) 3A , 3A

(4) 1A, 1A

21) In the Wheatstone bridge shown below, in order to balance the bridge, we must have :-



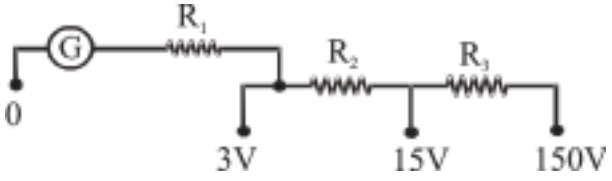
(1) $R_1 = 3\Omega$; $R_2 = 3\Omega$

(2) $R_1 = 6\Omega$; $R_2 = 15\Omega$

(3) $R_1 = 1.5\Omega$; R_2 = any finite value

(4) $R_1 = 3\Omega$; R_2 = any finite value

22) A voltmeter of variable ranges 3V, 15V, 150V is to be designed by connecting resistances R_1, R_2, R_3 in series with a galvanometer of resistance G = 20Ω, as shwon in figure. The galvanometer gives full scale deflection when a current of 1mA pass through its coil. Then, the resistances R_1 , R_2 and R_3 (in kilo ohms) should be, respectively :-

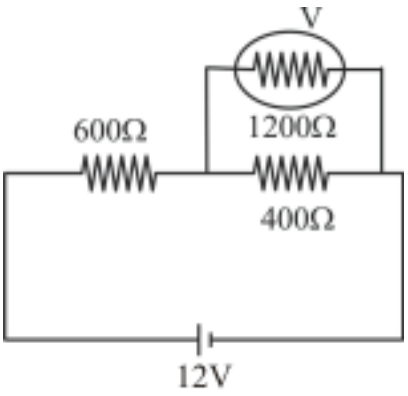


(1) 3, 12, 138

(2) 2.98, 12, 138

(3) 2.98, 14.98, 149.98

(4) 2.98, 12, 135



23) Find the error in reading of voltmeter

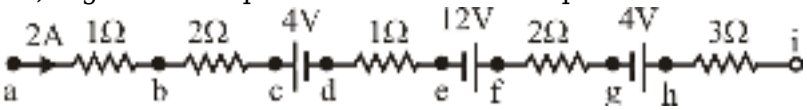
(1) 16.67%

(2) 8%

(3) 4%

(4) 5%

24) Figure shows part of a circuit. Which point have the same potential as that of point a ?



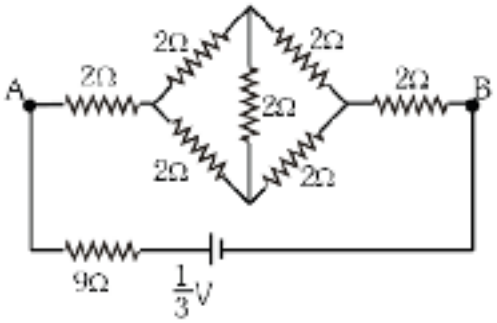
(1) f, h

(2) d, i

(3) f, i

(4) d, h

25) For the given circuit find out the–



(a) total resistance of the circuit across battery terminals.
(b) potential difference between A and B.

(1) 18/5 ohm, 1/5 Volt

(2) 15 ohm, 2/15 Volt

(3) 18 ohm, 2/15 Volt

(4) 15 ohm, 1/5 Volt

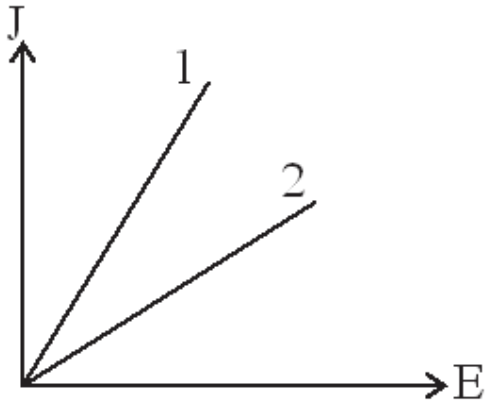
26) The resistance of platinum wire at 0°C is 2Ω and 6.8 Ω at 80°C. The temperature coefficient of resistance of the wire is :

(1) $3 \times 10^{-3}\,°C^{-1}$

(2) $3 \times 10^{-2}\,°C^{-1}$

(3) $3 \times 10^{-1}\,°C^{-1}$

(4) $3 \times 10^{-4}\,°C^{-1}$

27) 

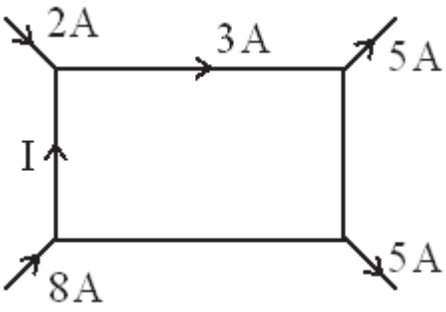
The correct relation between conductivity of wire 1 and wire 2 is :

(1) $\sigma_1 = \sigma_2$

(2) $\sigma_1 > \sigma_2$

(3) $\sigma_1 < \sigma_2$

(4) None

28) Calculate I . 

(1) 7A

(2) 2 A

(3) 5A

(4) 1A

29) A copper wire stretched so as to make it 10% longer. The percentage increase in the resistance of the wire is -

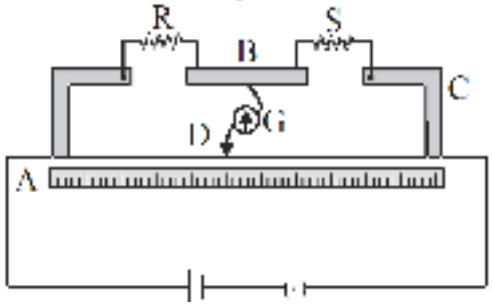
(1) 20%

(2) 21%

(3) 10%

(4) 50%

30) Circuit diagram of metre bridge is shown in figure. The null point is found at a distance of 40 cm from A. If now a resistance of 12Ω is connected in parallel with S, the null point occurs at 64 cm.



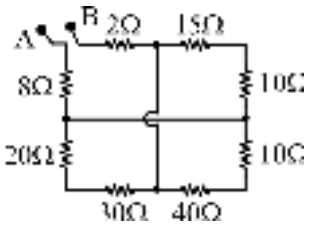
Determine the value of R and S.

(1) $\frac{40}{3}\Omega,\ 20\Omega$

(2) $\frac{10}{3}\Omega,\ 10\Omega$

(3) $11\Omega,\ 22\Omega$

(4) $\frac{80}{3}\Omega,\ 40\Omega$



31) The equivalent resistance between point A and B is

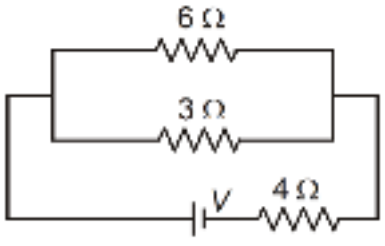
(1) $\frac{65}{2}\Omega$

(2) $\frac{45}{2}\Omega$

(3) $\frac{5}{2}\Omega$

(4) $\frac{91}{2}\Omega$

32) If power dissipated in the 6 Ω resistor in the circuit shown in figure is 30 watt, then the power



dissipated across the 3 Ω resistor is :

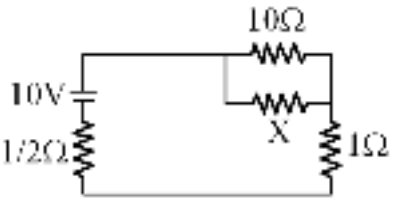
(1) 15 watt

(2) 30 watt

(3) 40 watt

(4) 60 watt

33) In given circuit power generated in resistance 1Ω will be maximum for x equal to :

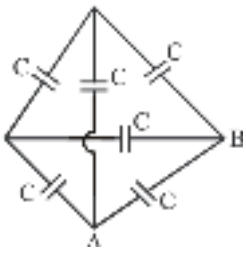


(1) 1Ω

(2) 2Ω

(3) 2/3Ω

(4) 0Ω



34) Find the equivalant capacitance in the given figure between A and B points ?

(1) C

(2) 2C

(3) C/2

(4) 3C

35) A parallel plate capacitor with air between plates has a capacitance of 8μF. What will be capacitance if distance between plates is reduced by half, and the space between them is filled with a substance of dielectric constant 6 ?

(1) 96 μF

(2) 24μF

(3) 50 μF

(4) 10 μF

36) When a slab of dielectric medium is placed between plates of a parallel plate capacitor which is connected with a battery, then the charge on plates in comparison with earlier charge :

(1) Is less

(2) Is same

(3) Is more

(4) Depends on the nature of the material inserted

37) In parallel plate capacitor the distance between plates is d = 0.1 mm. The medium between the plates is air. The maximum potential difference which can be applied to the capacitor is :- (dielectric strength of air = 3 MV/m)

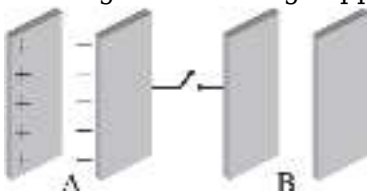
(1) 3×10^6 volt

(2) 300 volt

(3) 3×10^{10} volt

(4) infinite

38) Consider the situation shown in the figure. The capacitor A has a charge q on it whereas B is uncharged. The charge appearing on the capacitor B a long time after switch is closed is :-



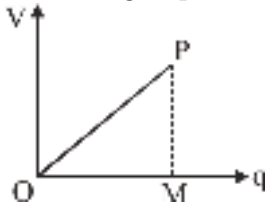
(1) Zero

(2) $\frac{q}{2}$

(3) q

(4) 2 q

39) Charge 'q' on a capacitor varies with voltage 'V' as shown. The area of ΔOPM represents:-



(1) capacitance

(2) capacitive reactance

(3) magnetic field between the plates

(4) energy stored in the capacitor

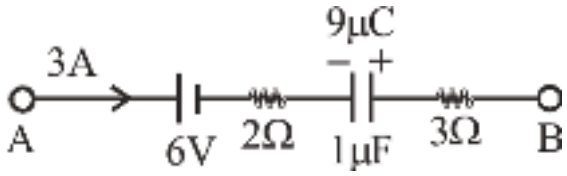
40) Two condensers of capacity 0.3 μF and 0.6 μF respectively are connected in series. The combination is connected across a potential of 6 volt. The ratio of energies stored by the condensers will be :-

(1) (1/2)

(2) 2

(3) (1/4)

(4) 4

41) 

Find $V_A - V_B$:-

(1) 12V

(2) –12V

(3) 6V

(4) –6V

42) Two flash light electric incandescent lamps, each requiring 3 amp at 1.5 V are placed in series

and connected to a 6 V cell. What resistance must be connected in series to operate them ?

(1) 22 Ω

(2) 2 Ω

(3) 6 Ω

(4) 1 Ω

43) **Assertion (A) :** Their will be electric field inside a current carrying wire :
Reason (R) : Net charge in a current carrying wire is zero.

(1) Both (A) & (R) are correct but (R) is the correct explanation of (A).

(2) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is not correct.

(4) (A) is not correct not but (R) is correct.

44) The charge flowing in a conductor varies with time as $Q = at - bt^2$. Which of the following is incorrect

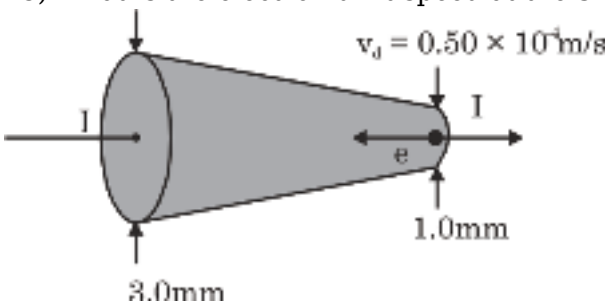
(1) The current decreases linearly with time

(2) The current increases linearly with time

(3) The current falls to zero at time $t = \frac{a}{2b}$

(4) The current changes at a rate of (– 2b)

45) What is the electron drift speed at the 3 mm diameter end (the left end) of the wire in figure.



(1) 0.055×10^{-4} m/s

(2) 4.5×10^{-4} m/s

(3) 0.055×10^{-3} m/s

(4) 4.5×10^{-3} m/s

CHEMISTRY

1) For the zero order reaction :
$A \rightarrow P$, $K = 10^{-2}$ (mol/litre) sec^{-1} If initial concentration of A is 0.3M , then find concentration of A left at 10 sec.

(1) 0 M

(2) 0.2 M

(3) 0.1 M

(4) 0.15 M

2) For a process A + B → product, the reaction is second order with respect to A and zero order with respect to B. When 1 mole each of A & B are taken in 1 litre vessel, the initial rate of disappearance of B is 1×10^{-2} mol/L-sec. The rate of reaction when 50% of the reactant have been converted to product would be -

(1) 1×10^{-2} mol/L-sec

(2) 2.5×10^{-3} mol/L-sec

(3) 5×10^{-2} mol/L-sec

(4) 5×10^{-3} mol/L-sec

3) The compound A decomposed (A → product) by first order kinetics. If the concentration of A dropped from 0.1 M to 0.05 M in 30 minutes, find the rate of reaction after 60 minutes, from start, in M min^{-1}

(1) $\frac{\ln 2}{1200}$

(2) $\frac{\ln 2}{600}$

(3) $\frac{\ln 2}{120}$

(4) $\frac{\ln 2}{60}$

4) Differential form of the rate equation is :-

$$\frac{dx}{dt} = k[P][Q]^{0.5}[R]^{0.5}$$

Which statement about the above equation is wrong ?

(1) Order with respect to P is one

(2) Total order of the reaction is two

(3) Order with respect to each of Q and R is 0.5

(4) Unit of specific reaction rate is $mol.L.s^{-1}$

5) The hydrogenation of vegetable ghee at 25°C reduces the pressure of H_2 from 2 atm to 1.2 atm in 50 min. Calculate the rate of reaction in terms of change of Molarity per second :-

(1) 3.27×10^{-2} Ms^{-1}

(2) 2.09×10^{-6} Ms^{-1}

(3) 1.09×10^{-5} Ms^{-1}

(4) 10.9×10^{-5} Ms^{-1}

6) The rate constant of a reaction is 0.0693 min^{-1}. Strating with 10 mol, the rate of the reaction after

10 min is :-

(1) 0.0693 mol min^{-1}

(2) 0.0693 × 2 mol min^{-1}

(3) 0.0693 × 5 mol min^{-1}

(4) 0.0693 × $(5)^2$ mol min^{-1}

7) The rate of disappearance of Q in the reaction : 2P + Q → 2R + 3S is 2 × 10^{-2} mol L^{-1} s^{-1}. Which of the following statement is not true :-

(1) $-\frac{d[P]}{dt} = 4 \times 10^{-2}$ mol L^{-1} s^{-1}

(2) $+\frac{d[S]}{dt} = 6 \times 10^{-2}$ mol L^{-1} s^{-1}

(3) $+\frac{1}{2} \cdot \frac{d[R]}{dt} = 2 \times 10^{-2}$ mol L^{-1} s^{-1}

(4) $+\frac{d[R]}{dt} = 2 \times 10^{-2}$ mol L^{-1} s^{-1}

8) For a reaction $2SO_2 + O_2 \rightarrow 2SO_3$ if rate of formation of SO_3 is 0.8 g/min. Then what will be rate of disappearance of SO_2

(1) 1.6 g/min

(2) 0.64 g/min

(3) 10^{-3} mol/min

(4) 10^{-4} g/min

9) For the reaction 2A → B + 3C ;
if $-\frac{d[A]}{dt} = k_1[A]^2, \frac{d[B]}{dt} = k_2[A]^2, \frac{d[C]}{dt} = k_3[A]^2$
the correct reaction between k_1, k_2 and k_3 is :-

(1) $k_1 = k_2 = k_3$

(2) $2k_1 = k_3 = 3k_2$

(3) $4k_1 = k_2 = 3k_2$

(4) $\frac{k_1}{2} = k_2 = \frac{k_3}{3}$

10) For a chemical reaction 2X + Y → Z, the rate of appearance of Z is 0.05 mol L^{-1} per min. The rate of disappearance of X will be :-

(1) 0.05 mol L^{-1} per hour

(2) 0.05 mol L^{-1} per min

(3) 0.1 mol L^{-1} min^{-1}

(4) 0.25 mol L^{-1} per min

11) Select the correct statement(s) regarding effect of catalyst on reaction :
(A) It is believed that catalyst provides an alternate pathway with lower activation energy.
(B) A catalyst does not alter Gibbs energy (ΔG) of reaction.
(C) It catalyses both spontaneous as well as non spontaneous reactions.
(D) Catalyst does not changes the equilibrium constant of reaction.
Choose the correct answer from the options given below :

(1) A, B and D

(2) C and D

(3) A, B and C

(4) C and B

12)

The decomposition of azo methane, at certain temperature according to the equation $(CH_3)_2N_{2(g)} \rightarrow C_2H_{5(g)} + N_{2(g)}$ is a first order reaction.

After 40 minutes from the start, the total pressure developed is found to be 350 mm Hg in place of initial pressure 200 mm Hg of azo methane. The value of rate constant k is :-

(1) $2.88 \times 10^{-4}\ s^{-1}$

(2) $1.25 \times 10^{-4}\ s^{-1}$

(3) $5.77 \times 10^{-4}\ s^{-1}$

(4) None of these

13)

In a reaction the following data has been observed:-

[A]/M	[B]/M	ROR/Ms^{-1}
0.10	0.10	0.40
0.20	x	0.80
y	0.20	1.60

If the reaction is 1st order with respect to both A & B then x & y respectively are :-

(1) 0.10, 0.10

(2) 0.20, 0.20

(3) 0.20, 0.10

(4) 0.10, 0.20

14) A reaction rate constant is given by :

$k = 1.2 \times 10^{14}\, e^{-\frac{2500}{RT}}\ sec^{-1}$. It means :-

(1) log k versus log T will give a straight line with a slope as 25000

(2) log k versus log T will give a straight line with a slope as –25000

(3) log k versus T will give a straight line with a slope as –25000

(4) log k versus 1/T will give a straight line

15) Match the column

Column I (Reduction process)		**Column II (Charge required)**	
(a)	1 mole of MnO_4^- to Mn^{2+}	(p)	193000 C
(b)	1 mole of $Cr_2O_7^{2-}$ to Cr^{3+}	(q)	289500 C
(c)	1 mole of Sn^{4+} to Sn^{2+}	(r)	482500 C
(d)	1 mole of Al^{3+} to Al	(s)	579000 C

(1) a → p, b → r, c → q, d → s

(2) a → r, b → s, c → q, d → p

(3) a → s, b → q, c → p, d → r

(4) a → r, b → s, c → p, d → q

16) Which of the following has least tendency to liberate H_2 from mineral acids ?

(1) Cu

(2) Mn

(3) Ni

(4) Zn

17) How much charge (in Faraday) is required to produce ?
(a) 36 g of Mg from molten $MgCl_2$ and
(b) 54 g of Al from molten Al_2O_3 respectively.

(1) (a) = 2 F ; (b) = 4 F

(2) (a) = 1.5 F ; (b) = 3 F

(3) (a) = 3 F ; (b) = 6 F

(4) (a) = 2 F ; (b) = 6 F

18) What will be the standard cell potential for the given cell ?
Zn/Zn^{2+} (0.1 M)||Cu^{2+} (0.01 M)|Cu
Given $E^0_{Zn^{2+}/Zn} = -0.76V,\ E^0_{Cu^{2+}/Cu} = +0.34V$

(1) –1.1 V

(2) 1.1 V

(3) –1.13 V

(4) 1.13 V

19) Match List-I with List-II :

List-I	**List-II**

(I)	Lead storage battery	(A)	Maximum efficiency
(II)	Mercury cell	(B)	Prevented by galvanisation
(III)	Fuel cell	(C)	Gives constant potential
(IV)	Rusting	(D)	Electrolyte is conc. H_2SO_4

Choose the correct answer from option given below :

(1) (I)-(D), (II)-(C), (III)-(B), (IV)-(A)

(2) (I)-(C), (II)-(D), (III)-(B), (IV)-(A)

(3) (I)-(D), (II)-(C), (III)-(A), (IV)-(B)

(4) (I)-(D), (II)-(A), (III)-(C), (IV)-(B)

20) A solution of $CuSO_4$ is electrolysed for 10 minutes with a current of 1.5A. The mass of copper deposited at cathode is :
[Atomic mass of copper = 63.5]

(1) 0.049 g

(2) 0.132 g

(3) 0.296 g

(4) 0.592 g

21) In the equation $\Lambda_m = \Lambda_m^o - b\sqrt{C}$, the value of b will be same for :

(1) NaCl and $CaCl_2$

(2) KCl and $MgSO_4$

(3) $CaCl_2$ and $MgSO_4$

(4) NaCl and KCl

22) Which of the following is strongest oxidising agent?
$E^o_{Cr_2O_7^{-2}/Cr^{+3}} = 1.33V; E^o_{Cl_2/Cl^-} = 1.36V$
$E^o_{MnO_4^-/Mn^{+2}} = 1.51V; E^o_{Cr^{+3}/Cr} = -0.74V$

(1) $Cr_2O_7^{-2}$

(2) Cl_2

(3) MnO_4^-

(4) Cr^{+3}

23) The value of Λ^∞ for NH_4Cl, NaOH and NaCl are respectively 150, 250 and 130 $ohm^{-1}cm^2eq^{-1}$. The value of Λ^∞ of NH_4OH is :-

(1) 270

(2) 280

(3) 400

(4) 380

24)

In an electrolytic cell, current flows

(1) from cathode to anode in outer circuit
(2) from anode to cathode outside the cell
(3) from cathode to anode inside the cell
(4) none of the above

25) Electrolysis of an aqueous solution of $CuSO_4$ using inert electrode will cause :-

(1) An increase in pH
(2) A decrease in pH
(3) Either decrease or increase in pH
(4) no change in pH

26) If hydrogen electrode is dipped into solution of pH = 3 and pH = 6 and salt bridge is connected, the emf of resulting cell is

(1) 0.177 V
(2) 0.3 V
(3) 0.052 V
(4) 0.104 V

27) The efficiency of a fuel cell is given by

(1) $\frac{\Delta H}{\Delta G}$
(2) $\frac{\Delta G}{\Delta S}$
(3) $\left|\frac{\Delta G}{\Delta H}\right|$
(4) $\frac{\Delta S}{\Delta G}$

28) Which of the following statement is incorrect?

(1) During electrolysis of dil. H_2SO_4 solution, water will be reduced at cathode.
(2) During electrolysis of dil. H_2SO_4 solution, water will be oxidised at anode
(3) During electrolysis of conc. H_2SO_4 solution, water will be reduced at cathode.
(4) During electrolysis of conc. H_2SO_4 solution, sulphate ions (SO_4^{-2}) will be oxidised at anode

29) Two liquid A and B have vapour pressure in the ratio $P_A^0 : P_B^0 = 1 : 3$ at a certain temperature. If the ratio of mole fractions of A to B in the vapour phase is 4 : 3, then the mole fraction of B in the solution at the same temperature is :

(1) $\frac{1}{5}$

(2) $\frac{2}{3}$

(3) $\frac{4}{5}$

(4) $\frac{1}{4}$

30) For 1% (w/v) H_2SO_4 solution find it's normality (approx)

(1) 0.1

(2) 0.2

(3) 0.4

(4) 0.02

31) Ratio of solubilities of gases N_2 & O_2 in water (as mole of gas in large volume of water) from air at 25°C & 1 atm will be, if air is 20 % by mole of O_2 & 80% by mole of N_2

Given : K_H (N_2) = 2×10^4 atm

K_H (O_2) = 10^4 atm

(1) 8 : 1

(2) 1 : 8

(3) 2 : 1

(4) 1 : 2

32) The mass of a non-volatile, non-electrolyte solute (molar mass = 50 g mol^{-1}) needed to be dissolved in 114 g octane to reduce its vapour pressure by 75%, is :-

(1) 50 g

(2) 37.5 g

(3) 75 g

(4) 150 g

33) Henry's constant (in kbar) for four gases α, β, γ and δ in water at 298 K is given below :

	α	β	γ	δ
K_H	50	2	2×10^{5}	0.5

(density of water = 10^3 kg m^{-3} at 298 K)

This table implies that :

(1) The pressure of a 55.5 molal solution of γ is 1 bar

(2) The pressure of a 55.5 molal solution of δ is 250 bar

(3) Solubility of γ at 308 K is higher than at 298 K

(4) α has the highest solubility in water at a given pressure

34) A 0.001 molal solution of $[Pt(NH_3)_4Cl_4]$ in water had a freezing point depression of 0.0054°C. K_f for water is 1.80 K kg mol^{-1}, the correct formulation for the above molecule is :

(1) $[Pt(NH_3)_4Cl_3]Cl$

(2) $[Pt(NH3)_4Cl_2]Cl_2$

(3) $Pt(NH_3)_4Cl]Cl_3$

(4) $[Pt(NH_3)_4Cl_4]$

35) An aqueous solution containing 5% by weight of urea and 10% by weight of glucose. Freezing point of solution is :
[K_f for H_2O is 1.86 K mol^{-1} kg]

(1) 2.78° C

(2) –3.04° C

(3) –5.96° C

(4) 5.96° C

36) Volume strength of H_2O_2 solution represents :-

(1) Volume of H_2O_2 solution

(2) Moles of H_2O_2 solution

(3) Volume of O_2 in litre obtained from 1 litre H_2O_2 solution at STP

(4) Volume of H_2 gas obtained at NTP

37) Which of the following solutions will have highest osmotic pressure :-

(1) 1 % Glucose in water

(2) 1 % Sucrose in water

(3) 1 % $CaCl_2$ in water

(4) 1 % urea in water

38) Two liquids A and B form ideal solution at 300 K. The vapour pressure of a solution containing one mole of A and four mole of B is 560 mm of Hg. At the same temperature if one mole of B is taken out from the solution, the vapour pressure of solution decreases by 10 mm of Hg. Vapour pressures of A and B in pure state will be-

(1) 330, 550

(2) 200, 300

(3) 400, 600

(4) 800, 1200

39) What will be the freezing point of 0.2 molal aqueous solution of $MgBr_2$? If salt shows 60% dissociation in solution and K_f for water is1.86 KKg mol^{-1}.

(1) –0.59 °C

(2) –0.67 °C

(3) –0.82 °C

(4) –1.04 °C

40) The vapour pressure of pure water is 92.5 mm at 300K. The vapour pressure of 1 molal solution of X in water is :-

(1) 180 mm

(2) 90.86 mm

(3) 45.4 mm

(4) Unpredictable because molar mass is not given

41) 0.4 m aqueous solution of monoprotic acid HX has elevation in boiling point equal to 0.3°.Calculate the percentage of dissociation of the acid.(K_b = 0.5 K m^{-1})

(1) 50%

(2) 40%

(3) 20%

(4) 100%

42)

Column-I (various solutions)		**Column-II (Their freezing point)**	
a	0.1M $BaCl_2$ solution	p	271 K
b	0.1M NaCl solution	q	270 K
c	0.1M K_3 $[Fe(CN)_6]$ solution	r	268 K
d	0.1M Al_2 $(SO_4)_3$ solution	s	269 K

Given : Freezing point of 0.1 M sucrose
Solution = 272 K and freezing point of water = 273 K:-
Which of the following option show correct matches? (assume, molarity=molality)

(1) (a–r), (b–p), (c–s), (d–q)

(2) (a–q), (b–p), (c–s), (d–r)

(3) (a–q), (b–s), (c–p), (d–r)

(4) (a–s), (b–q), (c–r), (d–q)

43) The heat of solution of anhydrous $CuSO_4$ and $CuSO_4.5H_2O$ are –15.89 and 2.80 kCal mol^{-1} respectively what will be the heat of hydration of $CuSO_4$:

(1) –18.69 kCal

(2) 18.69 kCal

(3) –28.96 kCal

(4) 28.96 kCal

44) For a reaction of iodide ion with hydrogen peroxide at room temperature what will be the order of reaction concerning iodide ion :

(1) Second order reaction

(2) Zero order reaction

(3) Third order reaction

(4) First order reaction

45) The mixture that forms maximum boiling azeotrope is :

(1) Water + Nitric acid

(2) Ethanol + Water

(3) Acetone + Carbon disulphide

(4) Heptane + Hexane

BIOLOGY

1) **Statement-I :** Prop roots are present in all *Ficus* species.
Statement-II : Fibrous roots are present in monocot plants.
Options :-

(1) Statement-I is correct and statement-II is incorrect.

(2) Statement-I and statement-II both are correct.

(3) Statement-I is incorrect and statement-II is correct.

(4) Statement-I and statement-II both are incorrect.

2) Match the **List-I** with **List-II**

List-I (Type of Aestivation)		**List-II (Example)**	
A.	Twisted	I.	Bean
B.	Valvate	II.	Gulmohar
C.	Vexillary	III.	Cotton
D.	Imbricate	IV.	*Calotropis*

Choose the correct answer from options given below:

(1) A-III, B-IV, C-II, D-I

(2) A-III, B-IV, C-I, D-II

(3) A-II, B-IV, C-I, D-III

(4) A-III, B-I, C-IV, D-II

3) Match the column :-

(a)	Meristematic region	(i)	Thimble like
(b)	Root cap	(ii)	Water absorption
(c)	Elongation region	(iii)	Very small, dense protoplasm
(d)	Maturation region	(iv)	Growth of root in length

	a	b	c	d
(1)	iii	i	iv	ii
(2)	ii	iv	iii	i
(3)	iv	iii	ii	i
(4)	i	ii	iii	iv

(1) 1

(2) 2

(3) 3

(4) 4

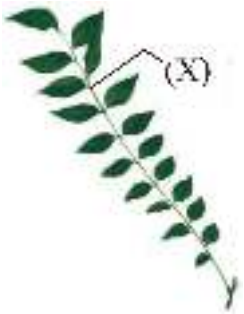
4) Tendrils present in grapevine and gourds develops from ?

(1) Leaf

(2) Axillary Bud

(3) Leaflet

(4) Stipule



5) What is the type of leaf and 'X' in the given structure below :-

(1) Pinnately compound, Rachis

(2) Pinnately compound, Petiole

(3) Palmately compound, Rachis

(4) Palmately compound, Petiole

6) In __A__ type of inflorescence the main axis __B__ to grow.
Fill in the blanks using the options given below :

(1) A-Cymose, B-Continuous

(2) A-Recemose, B-Discontinuous

(3) A-Spike, B-Discontinuous

(4) A-Recemose, B-Continuous

7) Given below are two statements : one is labelled as assertion (A) and the other is labelled as reason (R).
Assertion : A flower is defined as modified shoot where in the shoot apical meristem changes to floral meristem.
Reason : Node of the shoot gets condensed to produce different floral appendages laterally at successive internode instead of leaves.

In the light of the above statement, choose the correct answer from the option given below:

(1) Both **Assertion** and **Reason** are true but **Reason** is not the correct explanation of **Assertion**.
(2) **Assertion** is true but **Reason** is false.
(3) **Assertion** is false but **Reason** is true.
(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

8) In which of the following plants diadelphous stamen is found.

(1) China rose
(2) Pea
(3) *Citrus*
(4) *Salvia*

9) gymnosperms lack ___A__ in their xylem. __B__ is absent in most of the monocotyledons.

	(A)	(B)
(1)	Vessels	Phloem parenchyma
(2)	Vessels	Xylem parenchyma
(3)	Tracheids	Phloem parenchyma
(4)	Tracheids	Xylem parenchyma

(1) 1
(2) 2
(3) 3
(4) 4

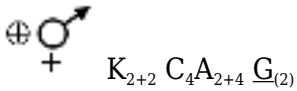
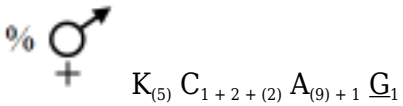
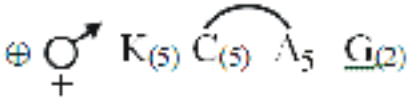
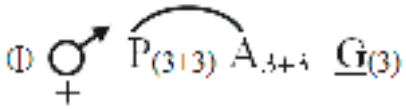
10) **Statement-I :-** Floral diagram provide information about the number of parts of a flower & their arrangement.
Statement-II :- Floral formula show cohesion & adhesion.
In the light of above statements. Choose the correct answer from the option given below :-

(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is correct but Statement-II is incorrect.
(3) Statement-I is incorrect but Statement-II is correct.
(4) Both Statement-I and Statement-II are correct.

11) Which of the following statement is correct?

(1) In a flower of onion stamens are six and epipetalous
(2) In a flower of tomato stamens are five and epitepalous
(3) In a flower of pea stamens are ten and monoadelphous
(4) In a flower of mustard stamens are six and variation in length of filaments of stamens

12) Match the column-I with column-II and select the correct option.

Column-I		Column-II	
(A)		I.	*Petunia*
(B)		II.	Lupin
(C)		III	Tulip
(D)		IV	Mustard

(1) A-I, B-II, C-IV, D-III

(2) A-II, B-I, C-IV, D-III

(3) A-IV, B-I, C-II, D-III

(4) A-IV, B-II, C-I, D-III

13) Select correct option for A, B, C and D.

(a) The floral formula for sun flower is 

(b) The outer covering of endosperm in a monocotyledons seed, separate the embryo by a proteinuous layer called ...B...

(c) The largest posterior petal in a pea flower is known as ...C...

(d) The ...D... in Australian acacia plant expand, become green and synthesise food.

	A	B	C	D
(1)	$K_{(5)}$	Aleurone layer	Keel	Petiole
(2)	K_P	Seed coat	Standard	Lamina
(3)	K_P	Aleurone layer	Standard	Petiole
(4)	K_5	Seed coat	Keel	Lamina

(1) 1

(2) 2

(3) 3

(4) 4

14) Pea, Gulmohar, Bean, *Cassia*, Mustard, Chilli, *Sesbania*, Lupin, Muliathi have many plants have zygomorphic flower ?

(1) Six

(2) Seven

(3) Four

(4) Eight

15) **Assertion (A) :-** Strawberry is an example of false fruit.
Reason (R) :- In strawberry, thalamus contributes to fruit formation.
In light of above statements, choose the most appropriate answer from option given below :-

(1) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(2) Assertion is true but Reason is false.
(3) Assertion is false but Reason is true.
(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

16) Members of sunflower family and grass family are similar in having of :

(1) Epigynous flower
(2) Basal placentation
(3) Inferior ovary
(4) All

17) Conjoint type of vascular bundles are common in :-

(1) Only stems
(2) Only leaves
(3) Only roots
(4) Both stems and leaves

18) Which of the following primary permanent tissues provides mechanical support to young dicotyledonous stem and petiole of leaf ?

(1) Chlorenchyma
(2) Collenchyma
(3) Sclerenchyma
(4) Xylem

19) Similarity in trichome and root hair is :-

(1) Both are epidermal out growth
(2) Both are always unbranched
(3) Both prevent transpiration
(4) Both are secretory

20) Monocot root differs from dicot root in having ?

(1) Open vascular bundles
(2) Scattered vascular bundles
(3) Well developed pith
(4) Radially arranged vascular bundles

21) In dicot roots, the cells of which region show Casparian strips ?

(1) Cambium

(2) Endodermis

(3) Pericycle

(4) Hypodermis

22) **Assertion (A) :** Moncotyledonous roots do not undergo any secondary growth.
Reason (R) : Vascular bundles are conjoint and closed in monocotyledonous root.

(1) Both **Assertion** and **Reason** are true but **Reason** is not the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

23) How many of the following statements are correct ?
(I) In leaves, the ground tissue consists of thin walled chloroplast containing cells and is called mesophyll.
(II) Radial arrangement of vascular bundles is found in stem.
(III) Conjoint type of vascular bundles are found in leaves.
(IV) In dicotyledonous stems, cambium is present between phloem and xylem.

(1) 1

(2) 2

(3) 3

(4) 4

24) **Statement-I :** The veins vary in thickness in the reticulate venation of the dicot leaves.
Statement-II : The parallel venation in monocot leaves is reflected in the near similar is sizes of vascular bundles except in main veins.

(1) Both statement-I and statement-II are incorrect.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect but statement-II is correct.

(4) Both statement-I and statement-II are correct.

25) Which of following statements are correct about dicot stem?
I. Hypodermis - Sclerenchymatous
II. Vascular bundle-Conjoint open with exarch protoxylem.
III. Endodermis rich in starch.
IV. Pith consist of parenchymatous cells with large intercellular spaces.

(1) I, II & III

(2) II & III

(3) II, III & IV

(4) III & IV

26) Which of the following statement is incorrect for dorsiventral leaf ?

(1) The epidermis cover both surface of leaf.

(2) The adaxial epidermis generally bear more stomata.

(3) The tissue between upper and lower epidermis is called the mesophyll.

(4) Vascular bundles are surrounded by bundle sheath cells.

27) In dicot stem arrange the structures in sequence from outside to inside.
(A) Hypodermis
(B) Protoxylem
(C) Epidermis
(D) Phloem
(E) Cambium
(F) Metaxylem
Choose the correct answer from the options given below :-

(1) C → E → A → B → D → F

(2) C → A → B → E → F → D

(3) C → A → D → E → F → B

(4) A → C → E → F → B → D

28) Examine the figure A-D given below and select the right option in which all the four figures A, B, C and D are identified correctly.

	A	B	C	D
				
1	Pollen grain tetrad	Mature embryo-sac	Ovule	Microspore
2	Pollen grain tetrad	Ovule	mature embryo sac	mature pollen grain
3	Megaspore tetrad	Ovule	mature embryo sac	mature pollen grain
4	Megaspore tetrad	Ovule	mature male gametophyte	mature pollen grain

(1) 1

(2) 2

(3) 3

(4) 4

29) Which of the following pair of statements is false?
(a) Pollen grains of Hydrophytes are aeroallergen.

(b) Mature pollen grains of angiosperms are two-celled.
(c) In *Citrus* and *Mangifera* some of the nucellar cells surrounding the embryo sac start dividing and protrude into the embryo sac and developing into embryos.
(d) Apomictic embryos are always haploid.

(1) (a) and (c)
(2) (b) and (d)
(3) (b) and (c)
(4) (a) and (d)

30) If number of chromosome in gamete of corn 12, what will be the number of chromosome in aleurone layer, coleoptile, synergid.

(1) 24,36,12
(2) 36,24,12
(3) 12,12,24
(4) 12,24,36

31) Identify the correct sequence of events taking place in angiosperms.
(A) Megasporogenesis
(B) Development of embryo
(C) Pollen-pistil interaction
(D) Development of endosperm
(E) Pollination
(F) Entry of pollen tube into the ovule
(G) Megagametogenesis

(1) A→E→G→C→F→B→D
(2) A→G→C→E→F→D→B
(3) A→G→E→C→F→D→B
(4) G→A→E→C→F→B→D

32) Identify the set of correct statements :
A. Water pollinated flowers are not very colourful and do not produce nectar.
B. The flowers of water hyacinth and water lily are pollinated by insects or wind.
C. In most of the water pollinated species the pollen grains are surrounded by a mucilaginous covering.
D. Pollen grains of some hydrophytes such as sea grass are long and ribbon like.
E. In *Vallisneria*, the male flowers reach the surface of water by the long stalk.
Choose the correct answer from the options given below :

(1) A, B, C, D only
(2) C, D and E only
(3) A, C , D and E only
(4) B, C, D and E only

33) Identify correct statement -
(A) Pollination by water is more common amongst abiotic pollinations.

(B) Pollination by water is limited to 30 genera.
(C) The flowers of water hyacinth are not pollinated by water.
(D) Wind pollinated flowers do not produce nectar.
(E) Among the animals birds are dominant biotic pollinating agents.

(1) C, D, E only
(2) A, B, C, D only
(3) B, C, D only
(4) B, C, D, E only

34) Read column-I, II & III and identify their correct match :

	Column-I		**Column-II**		**Column-III**
(A)	Anther	(P)	Megasporangia	(i)	Ovary
(B)	Pistil	(Q)	Primary endosperm cell	(ii)	Embryo
(C)	Syngamy	(R)	Microsporangia	(iii)	Endosperm
(D)	Polar nuclei	(S)	1-male gamete and egg cell	(iv)	Pollen sac

(1) A-R,iv; B-P,i; C-Q,iii; D-S,ii
(2) A-R,iv; B-P,i; C-S,ii; D-Q,iii
(3) A-P,i; B-R,iv; C-S,ii; D-S,iii
(4) A-P,i; B-S,iii; C-R,iv; D-S,iii

35) (i) Stamen is equivalent to microsporophyll
(ii) Whorls of androecium is called stamens.
(iii) The proximal end of filament is attached to the thalamus of the petals.
(iv) A typical angiosperm anther is bilobed with each lobe having four theca i.e they are dithecous.
(v) A typical anther is bilobed and tetrasporangiate
How many of the above statement is/are incorrect.

(1) Four
(2) One
(3) Three
(4) Two

36) Read column-I, II & III and identify their correct match :

	Column-I		**Column-II**		**Column-III**
(A)	Wind Pollination	(P)	Nector and non stricky	(i)	*Vallisnaria*
(B)	Water pollination	(Q)	Sticky pollen grain	(ii)	*Yucca*
(C)	Insect pollination	(R)	Non-sticky pollen grain	(iii)	Grasses

(D)	Insect or wind pollination	(S)	Ribbon like pollen grain	(iv)	Water hyacinth

(1) A-R,iii; B-S,i; C-P,iv; D-Q,ii

(2) A-S,i; B-R,iii; C-P,iv; D-Q,ii

(3) A-R,iii; B-S-i; C-Q,ii; D-P,iv

(4) A-P,iv; B-Q,ii; C-S,i; D-R,iii

37) (i) Synchrony in pollen release and stigma receptivity.
(ii) Cleistogamous flower produce assured seed set even in the absence of pollinators.
(iii) Cleistogamy is invariably autogamous as there is no chance of cross pollination.
(iv) Majority of flowering plants produce hermaphrodite flower.
(v) In cleistogamy flowers are open throughout the life.
How many statement are correct regarding adaptation for self pollination.

(1) 3

(2) 4

(3) 5

(4) 1

38) How many statement are not incorrect.
(a) Continued self pollination result in inbreeding depression.
(b) Geitonogamy is not true cross pollination.
(c) Geitonogamy is genetically self pollination.
(d) Allogamy is true cross pollination.
(e) Dioecious plant can show geitonogamy & Xenogamy.

(1) Three

(2) Two

(3) Four

(4) Five

39) Match the following-

	Column-I	**Column-II**	**Column-III**
A.	Endosperm	i (n)	a. Develops into embryo
B.	Zygote	ii (2n)	b. Guides the pole tube
C.	Filliform Apparatus	iii (3n)	c. Nutrition to developing embryos
D.	Antipodal cells		d. Degenerate post-fertilization

(1) A-ii-c B-ii-d C-i-a D-ii-b

(2) A-iii-c B-ii-a C-i-b D-ii-d

(3) A-iii-c B-i-a C-ii-b D-i-d

(4) A-i-c B-ii-b C-ii-d D-ii-a

40) Identify the correct match from column I, II, III

Column-I		Column-II		Column-III	
A.	Product of sungamy	(i)	Pollen grain	(a)	Male (♂) gametophyte
B.	Product of triple fusion	(ii)	Zygote	(b)	Embryo sac
C.	Product of microsporogausis	(iii)	PEN in PEC	(b)	Embryo
D.	Product of megasporogenesis	(vi)	Megaspore	(d)	Endosperm

(1) A-iii-d, B-iii-c, C-i-a, D-iv-b

(2) A-ii-c, B-iii-d, C-i-a, D-iv-b

(3) A-ii-c, B-iii-d, C-iv-b, D-i-a

(4) A-iii-d, B-ii-c, C-iv-b, D-i-a

41) **Statement-I :-** Pollens of Pollen banks can be used in crop breeding programmes.
Statement-II :- Nucellus have abundant reserve food material.

(1) Statement I correct and II incorrect

(2) Both statements are correct

(3) Both statements are incorrect

(4) Statement I incorrect and II is correct

42) Match the following and choose the correct option :

(a)	Epidermis	(i)	Single layer
(b)	Endothecium	(ii)	Multi-layer
(c)	Tapetum		
(d)	Middle layer		

(1) (a), (b)-i, (c), (d)-ii

(2) (a), (b), (c)-i, (d)-ii

(3) (a), (b), (d)-ii, (c)-i

(4) (a), (b)-ii, (c), (d)-i

43) How many given plants prevent autogamy but not geitonogamy?
(a) Papaya (b) Maize (c) Castor (d) Date palm

(1) One

(2) Two

(3) Three

(4) Four

44) Suppose a flower of china rose has 25 stamens, each microsporangium has 10 microspore mother cells. How many pollen grains occur in that flower?

(1) 500
(2) 1000
(3) 2000
(4) 100

45) In angiosperm plant one meiosis produces how many male gametes :-

(1) 4
(2) 2
(3) 1
(4) 8

46) The male sex accessory ducts include ……A……, …..B……, …….C….., and vas deferens. A, B and C in the given statements are :-

(1) A-Rete testis, B-Epididymis, C-Scrotum
(2) A-Spermatic cord, B-ureter, C-Scrotum
(3) A-Rete testis, B-Vasa efferentia, C-Epididymis
(4) A-Spermatic cord, B-Epididymis, C-Scrotum

47) Changes in GnRH pulse frequency in female is controlled by circulating level of :-

(1) Estrogen and Inhibin
(2) Progesterone and Relaxin
(3) Estrogen and progesterone
(4) Relaxin and inhibin

48) Secretion of which gland destroys the acidic medium of male urethra before release of semen ?

(1) Seminal vesicle
(2) Cowper's gland
(3) Prostate gland
(4) Pituitary gland

49) Which of the following pituitary hormone stimulate the release of female reproductive hormones?

(1) GnRH
(2) Estrogen
(3) ACTH
(4) FSH and LH

50) Ovulation in human a female takes place on 17th day of first menstrual bleeding. The duration of menstrual cycle in that female should be of:

(1) 05 Days

(2) 19 Days

(3) 28 Days

(4) 31 Days

51) **Assertion:** Mammary glands of the female undergo differentiation during pregnancy.
Reason: Hormonal changes during pregnancy prepares the mammary glands to produce milk for child care.

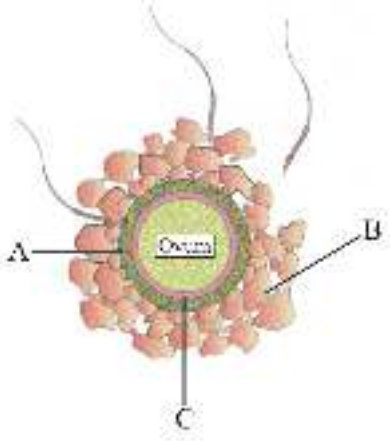
(1) Both assertion and reason are true and reason is correct explanation of assertion.

(2) Both assertion and reason are true but reason is not correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Both assertion and reason are false.

52) The given below diagram represents "ovum" surrounded by few sperms. Identify Labelled



structure A,B,C.

	A	B	C
(1)	Zona pellucida	Perivitelline space	Corona radiata
(2)	Zona pellucida	Corona radiata	Perivitelline space
(3)	Corona radiata	Perivitelline space	Zona pellucida
(4)	Corona radiata	Zona pellucida	Perivitelline space

(1) 1

(2) 2

(3) 3

(4) 4

53) Which of the following is incorrect?

(1) Testis is a primary sex organ in which sperms are produced

(2) Ovary is a primary sex organ in which ova are produced

(3) Vas deferens is a secondary sex organ which transports sperms

(4) Breast is a secondary sex organ which nourishes the infant

54) Match list I with list II

List-I		List-II	
A.	Embryo's heart is formed	(i)	By the end of the second month of pregnancy
B.	Foetus develops limbs and digits	(ii)	During the five month of pregnancy
C.	Major organ systems are formed	(iii)	By the end of 24 weeks
D.	Appearance of hairs on head	(iv)	After one month of pregnancy
E.	Formation of eyelashes	(v)	By the end of 12 weeks

Choose the option with all correct matches:

(1) A-(iv), B-(v), C-(i), D-(ii), E-(iii)

(2) A-(iv), B-(iii), C-(ii), D-(i), E-(v)

(3) A-(i), B-(iii), C-(ii), D-(v), E-(iv)

(4) A-(iv), B-(i), C-(v), D-(ii), E-(iii)

55) Match List-I with List-II

	List-I		**List-II**
(A)	Mons pubis	(I)	Tiny finger like structure
(B)	Labia majora	(II)	Paired fold of tissue
(C)	Labia minora	(III)	Fleshy fold of tissue
(D)	Clitoris	(IV)	Cushion of fatty tissue

Choose the correct answer from the options given below :

(1) A-IV, B-III, C-I, D-II

(2) A-IV, B-III, C-II, D-I

(3) A-III, B-IV, C-II, D-I

(4) A-III, B-II, C-I, D-IV

56) These are some events associated with reproduction and embryonic development. Arrange these from first to last and select the correct answer.

(a) Starts the formation of corpus luteum

(b) Cleavage

(c) Fertilisation

(d) Ovulation

(e) Implantation

(f) Formation of Graafian follicle

(g) Blastocyst
(h) Morula
(i) Zygote

(1) f, d, a, c, i, b, h, g, e
(2) f, d, i, c, a, b, h, g, e
(3) a, f, d, c, b, i, h, g, e
(4) i, h, g, f, d, c, b, e, a

57) What is not true about oogenesis in human females?

(1) The oogenesis starts in females before the birth of female herself.
(2) The complete oogenesis process produces four haploid ova.
(3) Second meiosis in secondary oocyte completes during fertilization.
(4) Female gamete releases from ovary in secondary oocyte stage.

58) Which of the following statements are correct?
(a) Fertilisation can occur only if the ovum and sperm are transported simultaneously to the ampulla.
(b) By the end of the second month of pregnancy the first movements of foetus can be observed.
(c) On the 14th day of menstrual cycle, ovulation takes place.
(d) Lack of menstruation is always an indication of pregnancy.
Options:

(1) a and c
(2) b and c
(3) c and d
(4) a and b

59) Which is not likely to be observed under the influence of estrogen hormone in females?

(1) Enlargement of breasts
(2) High pitch sound
(3) Development of pubic hairs
(4) Ejection of milk

60) Which hormone in females is secreted only during the pregnancy?

(1) Estrogen
(2) Progesterone
(3) Human Chorionic Gonadotropin
(4) Oxytocin

61) Which structure prevents the embryo from ectopic pregnancy?

(1) Corona radiata
(2) Zona pellucida

(3) Fertilisation membrane

(4) Graffian follicle

62) Which hormone maintains the pregnancy and makes the wall of uterus adhesive for implantation?

(1) FSH

(2) LH

(3) Estrogen

(4) Progesterone

63) The second meiotic division during oogenesis completes:

(1) Before birth

(2) Just after birth

(3) At the time of puberty

(4) During fertilization

64) Read the following Assertion (A) and Reason (R) and choose the correct option
Assertion (A): The clitoris is a tiny finger-like structure which lies at the upper junction of the two labia minora.
Reason (R): The labia minora are paired folds of tissue under the labia majora.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is incorrect

(4) Both (A) and (R) are incorrect

65) Match the following events with their resultant cells :

Column-1		**Column-2**	
A.	Meiosis-l	1.	Sperms
B.	Meiosis-II	2.	Spermatogonia
C.	Differentiation	3.	Ovum
D.	Mitosis	4.	First polar body

(1) A-4, B-1, C-3, D-2

(2) A-4, B-3, C-1, D-2

(3) A-4, B-1, C-2, D-3

(4) A-2, B-3, C-4, D-1

66) The secondary spermatocyte undergo the second meiotic division produce four equal spermatids. The spermatids are transformed into sperms by the process called spermiogenesis. After spermiogenesis sperm heads become embedded in the sertoli cells and are finally released into the lumen of seminiferous tubules by process called spermation. Given below are two statements about this method-

Statement-I : Spermatids are haploid cells that are formed after the second meiotic division.
Statement-II : Spermatids are diploid and undergoes mitosis to form sperm.

(1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect
(3) Statement II is correct and Statement I is incorrect
(4) Statement II is incorrect and Statement I is correct

67) **Statement-I :** Corpus luteum secretes progesterone which is essential for maintenance of the endometrium.
Statement-II : Fertilisation can only occur if the ovum and sperms are transported simultaneously to the ampullary region.

(1) Only statement-I is correct.
(2) Only statement-II is correct.
(3) Both statements are incorrect.
(4) Both statements are correct.

68) Which of the following statement is incorrect w.r.t. fallopian tubes ?

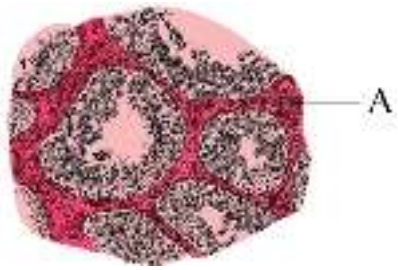
(1) Each ovary is covered by a thin epithelium which encloses the ovarian stroma.
(2) The last part of the oviduct, isthmus has a wide lumen and it joins the uterus.
(3) The endometrium undergoes cyclical changes during menstrual cycle.
(4) The labia majora are fleshy folds of tissue, which surround the vaginal opening.

69) The genetic material of a sperm is found in its:

(1) Head
(2) Middle part
(3) Neck
(4) Tail

70) How many sperms are produced by each secondary spermatocyte?

(1) 1
(2) 2
(3) 4
(4) 8



71) What is the function of structure A in the given diagram?

(1) Secretion of estrogen

(2) Formation of spermatozoa

(3) Secretion of testosterone

(4) Nutrition of spermatozoa

72) Which is not an accessory reproductive gland in human male reproductive system?

(1) Seminal vesicle

(2) Prostate gland

(3) Bartholin gland

(4) Cowper's gland

73) Given below are two statements are is labelled as Assertion and another is labelled as Reason
Assertion : In testis, the immature male germ cells (spermatogonia) produce sperms by spermatogenesis that begins at puberty.
Reason : A primary spermatocyte completes the first meiotic division (Reduction division) leading to formation of two equal haploid cell called secondary spermatocyte, which have only 23 chromosome each.
In the light of the above statements select the correct answer from the options given below :

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is correct but **Reason** is incorrect.

(4) **Assertion** is incorrect but **Reason** is correct.

74) **Statement-I :** Nearly 45 to 50 million MTP performed in a year in India which accounts to 1/5 th of the total number of conceived pregnancy in a year.
Statement-II : Government of India legalised MTP in 1971

(1) Both statement-I and statement-II are correct.

(2) Both statement-I and statement-II are incorrect.

(3) Statement-I is incorrect but statement-II is correct

(4) Statement-I is correct but statement-II is incorrect

75) Match the following :

A	Non- medicated IUDs	a	Multiload 375	i	Uterus unsuitable for implantation
B	Copper Releasing IUDs	b	Saheli	ii	Phagocytosis of sperm
C	Hormone Releasing IUDs	c	Lippes loop	iii	Suppress sperm motility
D	Weekly oral pill	d	LNG-20	iv	Anti-estrogenic activity

(1) A-b-i, B-c-ii, C-d-iii, D-a-iv

(2) A-c-iv, B-a-iii, C-b-i, D-d-ii

(3) A-c-ii, B-a-iii, C-d-i, D-b-iv

(4) A-a-iii, B-c-ii, C-d-i, D-b-iv

76) Which of the following contraceptive methods can prevent ovulation?

(1) CuT, Condoms
(2) Combination of estrogen and progesterone taken orally
(3) Diaphragms, cervical cap and vaults
(4) Coitus interruptus and lactational amenorrhoea

77) In test tube baby programme, the embryo more than 8 blastomeres is transferred to uterus, this is called as:

(1) ZIFT
(2) GIFT
(3) IUT
(4) ICSI

78) What is not true about SAHELI oral pill?

(1) It is weekly oral pill
(2) It is steroidal
(3) It is developed by scientists of CDRI
(4) It has high contraceptive value

79) Which of the following is a non-medicated IUD?

(1) Lippes loop
(2) Multiload 375
(3) Progestasert
(4) LNG-20

80) Match List-I with List-II

	List-I		**List-II**
(A)	In vitro fertilization	(I)	Gamete transfer
(B)	In vivo fertilization	(II)	Fertilization with in the body
(C)	ZIFT	(III)	Semen into uterus
(D)	GIFT	(IV)	Embryo upto 8 blastomere
(E)	IUT	(V)	Fertilization outside the body
(F)	IUI	(VI)	Embryo more than 8 blastomere

Choose the correct answer from the option given below:
(1) A-V, B-II, C-I, D-IV, E-III, F-VI
(2) A-V, B-I, C-II, D-III, E-VI, F-IV
(3) A-V, B-I, C-II, D-III, E-VI, F-IV

(4) A-V, B-II, C-IV, D-I, E-VI, F-III

81) Which of the following is not a part of IVF?

(1) ICSI

(2) ZIFT

(3) IUI

(4) IUT

82) Match the column-I with column-II and select the correct option about them.

	Column-I		**Column-II**
(a)	Lactational Amenorrhea	(i)	Male withdrawal his penis from vagina first before ejaculation
(b)	Coitus interruptus	(ii)	Suppress the sperm motility
(c)	Diaphragms	(iii)	Absence of menstrual cycle during the period of intense lactation
(d)	Cu-T	(iv)	Cover the cervex during coitus

(1) a-(i), b-(ii), c-(iii), d-(iv)

(2) a-(iii), b-(i), c-(iv), d-(ii)

(3) a-(iv), b-(iii), c-(ii), d-(i)

(4) a-(ii), b-(iv), c-(i), d-(iii)

83) Given below are two statements :
Statement-I : The early symptoms of sexually transmitted infections are minor and include itching, fluid discharge, slight pain and swelling etc.
Statement-II : Absence or less significant symptoms in the early stages of infection and the social stigma attached to STIs, deter the infected person from going for timely detection and proper treatment.
Select the correct answer from the options given below :-

(1) Statement-I and II both are correct.

(2) Statement-I and II both are incorrect.

(3) Statement-I is correct but II is incorrect.

(4) Statement-I is incorrect but II is correct.

84) Read the following statements and choose the correct option
Statement A: The Medical Termination of Pregnancy Act was enacted by the government of India in 2017.
Statement B: MTPs are considered relatively safe during the first trimester.

(1) Both the statements are correct

(2) Both the statements are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

85) Which of the following statement is incorrect?

(1) Inability to conceive or produce children even after 2 years of unprotected sexual cohabitation is called infertility.

(2) Embryos formed by in-vivo fertilisation can be used for embryo transfer.

(3) Legal adoption is one of the best methods for couples looking for parenthood.

(4) In artificial insemination, the semen collected from a healthy donor is artificially introduced into the vagina, it is called intra-uterine transfer.

86) Match the following-

	Column-I		**Column-II**
(i)	Surgical intervention	(a)	Inhibits ovulation
(ii)	Pills	(b)	Blocks gametes transport
(iii)	Saheli	(c)	Non steroidal pills
(iv)	IUD	(d)	To delay pregnancy

(1) (i-b), (ii-c), (iii-a), (iv-d)

(2) (i-a), (ii-b), (iii-c), (iv-d)

(3) (i-b), (ii-a), (iii-c), (iv-d)

(4) (i-c), (ii-b), (iii-a), (iv-d)

87) **Statement-1 : -** According to world health organization, reproductive health means a total well being in all aspects of reproduction.
Statement-2 : - Family planning programmes were initiated in 1951.

(1) Statement-1 is correct & statement-2 is incorrect

(2) Statement-2 is correct & statement-1 is incorrect

(3) Both statement 1 & 2 are incorrect.

(4) Both statement 1 & 2 are correct.

88) Except hepatitis-B, _____, and HIV infections, other diseases are completely curable if detected early and treated properly.

(1) Gonorrhoea

(2) Genital warts

(3) Genital herpes

(4) Syphilis

89) The hormonal pills have to be taken daily for a period of _____ starting preferably from the, first day of menstrual cycle.

(1) 7 days

(2) 14 days

(3) 21 days

(4) 28 days

90) Which one is a copper-releasing IUD?

(1) Lippes loop
(2) Multiload 375
(3) LNG-20
(4) Vaginal ring

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	1	2	2	2	4	3	1	4	2	2	1	3	1	3	2	2	3	2	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	4	1	1	2	2	2	4	2	1	2	4	4	2	1	3	2	1	4	2
Q.	41	42	43	44	45															
A.	1	4	2	2	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	2	1	4	3	3	4	2	4	3	1	3	4	4	4	1	3	2	3	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	3	1	1	2	1	3	3	1	2	3	4	2	2	2	3	3	3	3	2
Q.	86	87	88	89	90															
A.	1	2	1	4	1															

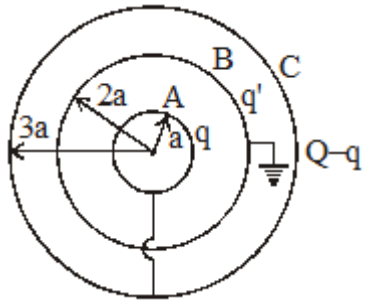
BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	2	1	2	1	4	2	2	1	4	4	4	3	2	4	2	4	2	1	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	2	3	4	4	2	3	2	4	2	3	1	3	2	4	3	2	1	2	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	2	2	3	4	3	3	2	4	4	1	2	4	4	2	1	2	1	4	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	4	4	2	2	4	4	2	1	2	3	3	2	3	3	2	3	2	1	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	2	1	4	4	3	4	3	3	2										

SOLUTIONS

PHYSICS

1) $V_B = 0$



$$\frac{kq}{2a} + \frac{kq'}{2a} + k\frac{(Q-q)}{3a} = 0 \quad \text{...(i)}$$

$V_A = V_C$

$$\frac{kq}{a} + \frac{kq'}{2a} + \frac{k(Q-q)}{3a} = \frac{k(q+q'+Q-q)}{3a} \quad \text{...(ii)}$$

from (i) & (ii)

$$q' = \frac{-8Q}{11}$$

2)

$$U_2 = U_1 + qV \Rightarrow V = \frac{U_2 - U_1}{q}$$

3)

Given $E_{in} = E_{out}$

$$\frac{kQ}{(2R-x)^2} = \frac{k(4Q)}{(2R+x)^2}$$

$$\frac{2R+x}{2R-x} = 2 \Rightarrow x = \frac{2R}{3}$$

4)

Max length of wire in the sphere

$$L = 2\sqrt{5^2-3^2} = 8m$$

Charge on wire is $q = \gamma L = \frac{8.85\ \mu C}{m} \times 8m$

$= 8.85\mu c \times 8\mu C$

Flux through the sphere $= \frac{q}{\epsilon_0} = \frac{8.85 \times 10^{-6}}{8.85 \times 10^{-12}}$

$$\phi = 8 \times 10^6 \frac{N-m^2}{C}$$

5) $V = \frac{1000}{x} + \frac{1500}{x^2} + \frac{500}{x^3}$

$E = -\frac{\partial V}{\partial x}\hat{i} - \frac{\partial V}{\partial y}\hat{j} - \frac{\partial V}{\partial z}\hat{k}$

$\frac{\partial V}{\partial x} = -\frac{1000}{x^2} - \frac{3000}{x^3} - \frac{1500}{x^4}$

$\frac{\partial V}{\partial y} = 0 \ \& \ \frac{\partial V}{\partial z} = 0$

$E(x,0,0) = \left(\frac{1000}{x^2} + \frac{3000}{x^3} + \frac{1500}{x^4}\right)\hat{i}\ V/m$

$E(1,0,0) = (1000 + 3000 + 1500)\hat{i}\ V/m$

$= 5500\hat{i}\ V/m$

6) $E = \frac{\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0} = \frac{\sigma}{\epsilon_0},$

$P = qd$

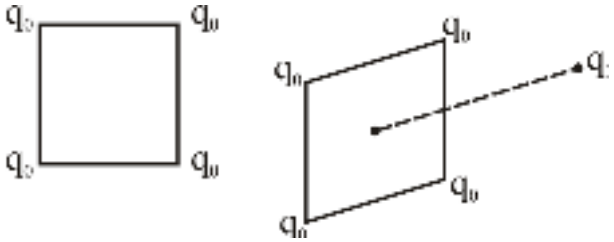
$F_{net} = 0,$

$\tau = PE\sin\theta$

$U = -PE\cos\theta$

7)

q_1 is in unstable equilibrium if q_0 & q_1 have same signs.

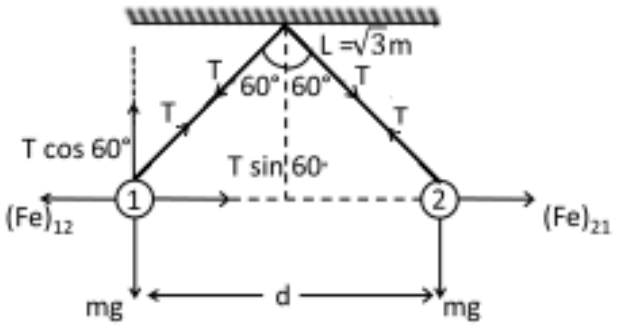


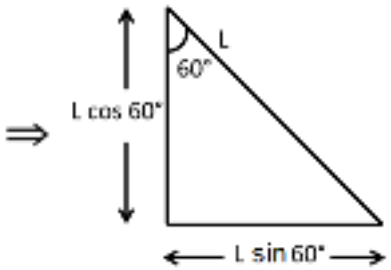
As charge moves toward the (+) ve z-axis or –ve z-axis, then net force on charge will be along the displace direction& would not come ever.

q_1 is in stable equilibrium if q_1 & q_0 have difference signes

In this case as charge moves away from centre then net force on the charge opposite to displace direction.

8)





$d = 2 \times L \sin 60°$

$d = 2 \times L \times \frac{\sqrt{3}}{2}$

$d = 2 \times \sqrt{3} \times \frac{\sqrt{3}}{2} = 3m$

$F_e = T \sin 60°$

$\frac{kQ^2}{d^2} = T \times \frac{\sqrt{3}}{2}$

$\frac{9 \times 10^9 \times \left(10 \times 10^{-6}\right)^2}{(3)^2} = T \times \frac{\sqrt{3}}{2}$

$0.1 = T \times \frac{\sqrt{3}}{2}$

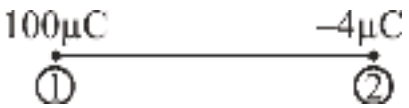
$T = \frac{2 \times 0.1}{\sqrt{3}} = \frac{0.2}{\sqrt{3}} N$

9)

$V = V_{\text{due to q}} + V_{\text{indue charge at inner Surface}} + V_{\text{indue charge at outer Surface}}$

$= \frac{kq}{d} + \left(-\frac{kq}{R_1}\right) + \left(\frac{kq}{R_2}\right)$

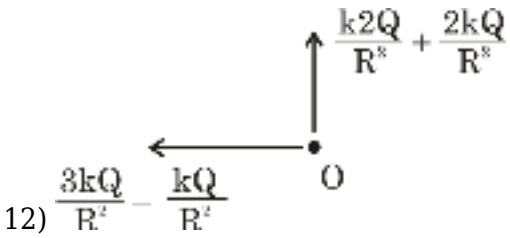
10)

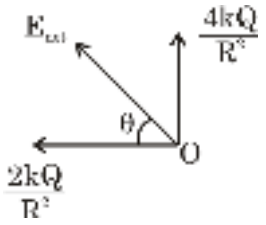


$\vec{F}_{21} = \frac{kq_1q_2}{|\vec{r}_{12}|^3}\vec{r}_{12}, \vec{r}_{12} = \vec{r}_2 - \vec{r}_1$

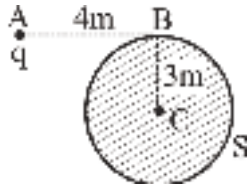
11) $\oint \vec{E} \cdot \vec{ds} = \frac{q_{in}}{\epsilon_0}$

$E \cdot 4\pi\left(\frac{R}{2}\right)^2 = \frac{\int_0^{R/2} \rho 4\pi r^2 dr}{\epsilon_0}$

12) 



$\tan\theta = 2$ $E_{net} = \frac{kQ}{R^2}\cdot\sqrt{20}\,N/C$

13) 

$\frac{kq}{5} = \frac{kq}{4} + V_{ind.}$

$V_{ind.} = -9\times10^3\left(\frac{2}{15}\right) = -0.45\ kV$

14) For the motion of electron along y–axis.

$d = \frac{1}{2}\frac{eE}{m}t^2$ ...(i) $\quad E = \frac{\sigma}{\varepsilon_0}$ ...(ii)

$t = \frac{\ell}{u}$ ....(iii)

From equation (i), (ii) & (iii)

$d = \frac{1}{2}\frac{e\sigma}{\varepsilon_0 m}\left(\frac{\ell}{u}\right)^2$; $\sigma = \frac{2d\varepsilon_0 mu^2}{e\ell^2}$

15) $q = \int_0^R \rho dV = \int_0^R \rho_0\left(\frac{5}{4}-\frac{r}{R}\right)4\pi r^2\,dr = \frac{2\pi\rho_0 R^3}{3}$

At surface

$V = \frac{1}{4\pi\epsilon_0}\frac{q}{R}, = \frac{1}{4\pi\epsilon_0 R}\times\frac{2\pi\rho_0 R^3}{3} = \frac{\rho_0 R^2}{6\epsilon_0}$

16) $\vec{\tau} = \vec{P}\times\vec{E} = 9\times10^{-6}\left(\hat{i}-\hat{j}+\hat{k}\right)\times0.20\times100\,\hat{i}$

$\vec{\tau} = 18\times10^{-5}(\hat{k}-\hat{j})$

$|\vec{\tau}| = 25.45\times10^{-5}N-m$

17) $\sigma \propto \frac{1}{r}$ and $E = \frac{\sigma}{\epsilon_0}$
r is radius of curvature.

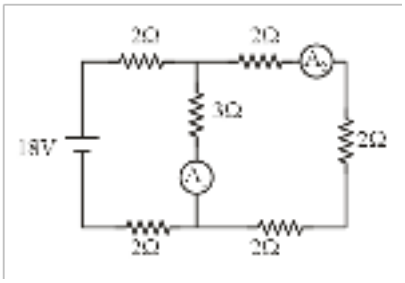
18) $V_+ = -(V-)$ at the axial position. PE of system is zero

19)

Potential difference across each resistance is zero hence current will be zero.

20)

Effective circuit is



$i = \frac{E_{net}}{R_{net}} = \frac{18}{6} = 3A$

$\frac{i_1}{i_2} = \frac{R_2}{R_1} = \frac{6}{3} = 2:1$

$i_1 = \frac{2}{3}(3A) = 2A$; $i_2 = \frac{1}{3}(3A) = 1A$

21)

For balance W.S.B

$\frac{P}{Q} = \frac{R}{S} \Rightarrow \frac{10}{R_1} = \frac{30}{9}$

$R_1 = 3\Omega$

22) $R_H = \frac{V}{I_g} - R_g$

$R_1 = \frac{3}{10^{-3}} - 20$

$= 2980\ \Omega = 2.98\ K\Omega$

$R_1 + R_2 = \frac{15}{10^{-3}} - 20 = 14,980\Omega$

$R_2 = 12\ k\Omega$

$R_1 + R_2 + R_3 = \frac{150}{10^{-3}} - 20$

$R_3 = 135\ k\Omega$

23) When resistance of voltmeter is ∞

$$V_1 = 12 \times \frac{400}{1000} = 4.8V$$

When resistance of voltmeter taken into account

$$V_2 = \frac{12 \times 300}{900} = 4V$$

$$\text{error} = \frac{4.8-4}{4.8} \times 100\% = \frac{0.8}{4.8} \times 100\% = \frac{50}{3}\%$$

$=16.67\%$

24) $V_A - 2 - 4 - 4 - 2 + 12 = V_f$

$V_A - V_f = 0 \quad \Rightarrow V_A = V_f$

$V_f - 4 + 4 = V_h \quad \Rightarrow V_f - V_h = 0$

$\Rightarrow V_f = V_h$

$\Rightarrow V_A = V_f = V_h$

25)

This is balance W.S.B So

$R_{eq} = 9 + 2 + 2 + 2 = 15\ \Omega$

$$I = \frac{E}{R_{eq}} = \frac{1}{3(15)}A$$

$$V_A - V_B = IR = \frac{1}{3(15)}(6)$$

$$= \frac{2}{15}V$$

26) $R_T = R_0[1 + \alpha(T - T_0)]$

$6.8 = 2[1 + \alpha(80 - 0)]$

$$\alpha = \frac{2.4}{80} = 0.03/°C = 3 \times 10^{-2}/°C$$

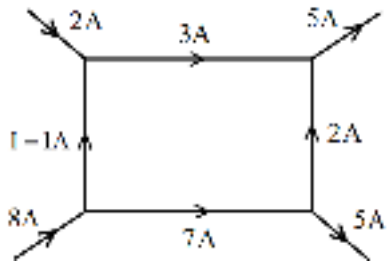
27)

$J = \sigma E$

$$\text{Slope} = \tan\theta = \sigma = \frac{J}{E}$$

28)

use K.C.L.



29) $L' = 1.1L$

$$R' = \rho\frac{(1.1L)}{\left(\frac{A}{1.1}\right)} = (1.1)^2\frac{\rho L}{A} = 1.21R$$

$$\% \text{ change} = \frac{R'-R}{R} \times 100 = 21\%$$

30) $\frac{40}{60} = \frac{R}{S}, \frac{2}{3} = \frac{R}{S}$ ...(1)

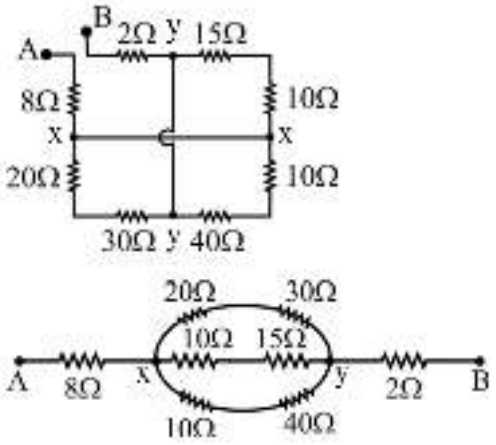
$$\frac{64}{36} = \frac{R(12+S)}{12S}$$

$$\frac{16}{9} = \frac{R(12+S)}{12S}$$ ...(2)

(1)/(2)

$S = 20\Omega,\ R = \frac{40}{3}\Omega$

31) Use Equipotential method
(समविभव बिन्दु तकनीक)



$$R_{eq} = 8 + \frac{25}{2} + 2 = \frac{45}{2}\Omega$$

32)

In parallel -
Voltage same

$$P = \frac{V^2}{R}$$

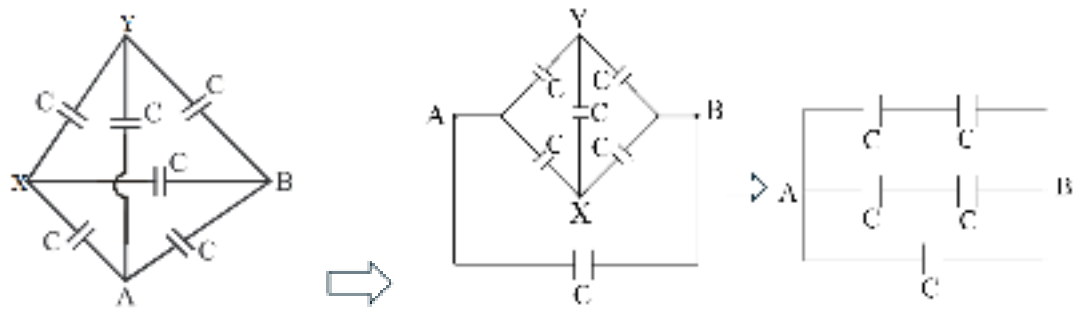
$$P \propto \frac{1}{R}$$

$$\frac{P_3}{P_6} = \frac{6}{3} \Rightarrow P_3 = \frac{6}{3}(30) = 60 \text{ W}$$

33) For power to be maximum on :
Fixed Resistance, variable resistance is zero.
⇒ Current is maximum.

⇒ Power is maximum.

34)



$$\frac{1}{C_{eq}} = \frac{1}{2C} + \frac{1}{2C} + \frac{1}{C}$$
$$C_{eq} = 2C$$

35)

$C \propto (K / d)$
$\Rightarrow C_2 = K \times (d_1 / d_2) \times C_1$
Given $C_1 = 8\ \mu F$, $K = 6$, and $d_2 = d_1/2$
$C_2 = K \times (d_1 / d_2) \times C_1$
$C_2 = 6 \times (d_1 / (d_1/2)) \times 8$
$C_2 = 6 \times 2 \times 8 = 96\ \mu F$

36)

$q = CV$
V = constant as battery is connected on insertion of dielectric C increases so q also increases.

37)

$V = Ed$
$V_{max} = E_{max} \times d$
$= 3 \times 10^6 \times 10^{-4}$
$= 3 \times 10^2 = 300$ Volt

38)

Charge on Inner Plates is bound, So $q_B = 0$

39)

area of $\Delta OPM = \frac{1}{2} \times q \times V$ = energy stored in the capacitor

40)

in series combination of capacitor ⇒ Q = constant and energy $E = \frac{Q^2}{2C}$

$\Rightarrow E \propto \frac{1}{C}$

$\Rightarrow \frac{E_1}{E_2} = \frac{C_2}{C_1} = \frac{2}{1}$

41)

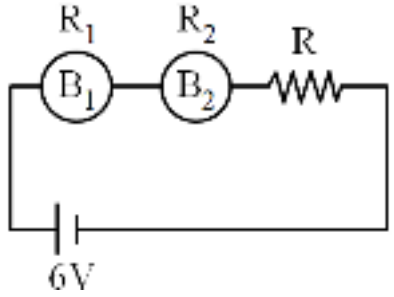
from KVL

$V_A - 6 - 6 + 9 - 9 = V_B$

$V_A - V_B = 12$ Volt

42) For series

$I = \frac{V}{R_{net}}$



$R_B = \frac{V}{I} = \frac{1.5}{3} = 0.5\ \Omega$

In series I → same

$I = \frac{V}{R_{eq}}$

$3 = \frac{6}{0.5 + 0.5 + R}$

$3 + 3R = 6 \Rightarrow R = 1\Omega$

43) Electric field inside a current carrying conductor is due to potential difference created by battery.

44) $I = \frac{dq}{dt} = \frac{d}{dt}\left(at - bt^2\right) = a - 2bt$

$I = a - 2bt = 0$

$\Rightarrow t = \frac{a}{2b}$

45)

Solution: $I = neAv_d$

$v_d \propto \frac{1}{A} \propto \frac{1}{r^2}$

$$\frac{v_{d_{left}}}{v_{d_{right}}} = \frac{r^2_{right}}{r^2_{left}} = \left(\frac{1}{3}\right)^2$$

$$v_{d\,left} = \frac{v_{d_{right}}}{9} = \frac{0.50 \times 10^{-4}}{9} m/s$$

$v_{d\,left} = 0.055 \times 10^{-4}$ m/s

CHEMISTRY

46)

$[A]_t = [A]_0 - Kt$
$= 0.3 - 10{-}2 \times 10$
$= 0.2$ M

47) **Question Explanation:** Find rate of reaction when 50% of the reactant have been converted to product.

Concept: This question is based on First order reaction.

Solution:
$r = K [A]^2 [B]^0$
$10^{-2} = K(1)^2 (1)^0 \Rightarrow K = 10^{-2}$
finally, $r = 10^{-2}(1/2)^2 (1/2)^0$
$= 2.5 \times 10^{-3}$

Final Answer: Option (2)

48)

First order reaction

$$0.1M \xrightarrow[30\ min]{I^{st} t_{1/2}} 0.05M \xrightarrow[30\ min]{2^{nd} t_{1/2}} 0.025M$$

$$t_{1/2} = \frac{\ell n 2}{k}$$

$$k = \frac{\ell n 2}{30}\ min^{-1}$$

rate of reaction after 60 min.
$r = k[A]_t$

$$r = \frac{\ell n 2}{30} \times (0.025)$$

$$r = \frac{\ell n 2}{1200}$$

49)

Question Explanation: We are given a differential rate law and must identify which statement about it is **wrong**.

Concept: Rate law

Solution:
Given rate law: $dx/dt = k\,[P]^1\,[Q]^{0.5}\,[R]^{0.5}$

A. Order w.r.t. P = 1
B. Order w.r.t. Q = 0.5
C. Order w.r.t. R = 0.5
D. Total order = 1 + 0.5 + 0.5 = **2**
Units of k (for 2nd-order overall): Rate unit = mol L^{-1} s^{-1}
Concentration^2 = (mol L^{-1})^2
k = (mol L^{-1} s^{-1}) / (mol^2 L^{-2}) = **L mol^{-1} s^{-1}**

A. Order with respect to P is one → **Correct**

B. Total order is two → **Correct**

C. Order w.r.t. Q and R is 0.5 each → **Correct**

D. Unit of rate constant is mol·L·s^{-1} → **Wrong** (correct unit = L·mol^{-1}·s^{-1})

Final Answer: Option 4

50)

ROR = $\frac{\Delta P}{\Delta t} = \frac{2-1.2}{50 \times 60}$ atm/sec
$\Delta P = \Delta CRT$
$\frac{\Delta C}{\Delta t} = \frac{\Delta P}{RT\Delta t} = \frac{0.8}{50 \times 60 \times 0.0821 \times 298}$
$= 1.09 \times 10^{-5}$ $Msec^{-1}$.

51)

Question Explanation:
The question describes a first-order reaction with a given rate constant and initial amount of reactant. We need to determine the rate of the reaction after a specific time.

Concept: First order Kinetics
For a first-order reaction, the rate law is: Rate = k[A]

Solution:
Given:

A. Rate constant (k) = 0.0693 min^{-1}
B. Initial amount (A_0) = 10 mol
C. Time (t) = 10 min

Since the units of the rate constant are min^{-1}, this is a first-order reaction.

As we know, for a first order reaction,

$t_{1/2} = \frac{0.693}{k} = \frac{0.693}{0.0639} = 10$ min

Reactant after 10 min = 5 mol

Rate $\left(\frac{dx}{dt}\right) = k[A] = 0.0639 \times 5$ mol min^{-1}

The rate of the reaction after 10 minutes is 0.0693×5 mol min^{-1}.

final answer; option 3.

52)

Question Explanation: Using the reaction stoichiometry, relate the given rate of disappearance of Q to rates of P, R, and S. Identify which statement does **not** satisfy stoichiometric rate relations.

Concept: Relation between rates of disappearance/appearance using stoichiometry.

Solution:

For the reaction: $2P + Q \rightarrow 2R + 3S$

Rates relate as: $-\frac{d[P]}{dt} = 2 \times -\frac{d[Q]}{dt}$, $\frac{d[R]}{dt} = 2 \times -\frac{d[Q]}{dt}$, $\frac{d[S]}{dt} = 3 \times -\frac{d[Q]}{dt}$

Given: $-\frac{d[Q]}{dt} = 2 \times 10^{-2}$ mol L^{-1} s^{-1}

A. $-\frac{d[P]}{dt} = 2 \times (2\times10^{-2}) = 4 \times 10^{-2} \rightarrow$ correct
B. $\frac{d[S]}{dt} = (3/1) \times (2\times10^{-2}) = 6 \times 10^{-2} \rightarrow$ correct
C. $(1/2)\frac{d[R]}{dt} = (1/2) \times 4\times10^{-2} = 2 \times 10^{-2} \rightarrow$ correct
D. $\frac{d[R]}{dt} = 2 \times (2\times10^{-2}) = 4 \times 10^{-2}$, but given value $= 2 \times 10^{-2} \rightarrow$ incorrect

Final Answer: Option 4

53) **Explain Question:**

To find out ROD of SO_2.

Concept:

ROA and ROD.

Solution:

$2SO_2 + O_2 \rightarrow 2SO_3$

$(ROA)SO_3 = 0.8$ g/min

$= \frac{0.8}{80}$ mol min^{-1}

$= 0.01$ mol min^{-1}

$\frac{(ROD)_{SO_2}}{2} = \frac{(ROA)_{SO_3}}{2}$

$(ROD)_{SO_2} = (ROA)_{SO_3}$

$= 0.01$ mol mm^{-1}

$= 0.01 \times 64$ g min^{-1}

$= 0.64$ g min^{-1}

Final answer: Option (2)

54) Rate of reaction = $-\frac{1}{2}\frac{d[A]}{dt}$

$= \frac{d[B]}{dt} = \frac{1}{3}\frac{d[C]}{dt}$

or $\frac{k_1[A]^2}{2} = \frac{k_2[A]^2}{1} = \frac{k_3[A]^3}{3}$

$\frac{k_1}{2} = k_2 = \frac{k_3}{3}$

55)

Question Explanation: Find the rate of disappearance of X using the rate of appearance of Z and the stoichiometric relation.

Concept: For the reaction 2X + Y → Z, disappearance and appearance rates are related by stoichiometry.

Solution:
Reaction: 2X + Y → Z
Rate relation: (1/2)(rate of disappearance of X) = (rate of appearance of Z)
Given rate of appearance of Z = 0.05 mol L^{-1} min^{-1}
Rate of disappearance of X = 2 × 0.05 = 0.1 mol L^{-1} min^{-1}

Final Answer: Option 3

56)

Catalyst does not change ΔG and ΔH and it does not change equilibrium constance (K) also.

57)

$(CH_3)_2N_{2(g)} \rightarrow C_2H_{5(g)} + N_{2(g)}$
200
200 – x x x
200 – x + x + x = 350
x = 150
$K = \frac{\ell}{t}\ell n\left(\frac{P_0}{Pt}\right) = \frac{1}{40 \times 60}\ell n\left(\frac{200}{50}\right)$
$= 5.77 \times 10^{-4}$ sec^{-1}

58)

$R = K[A]^1[B]^1$
Solve the eq^n. for value of R, [A] & [B].

59)

$k = 1.2 \times 10^{14} e^{\frac{-2500}{RT}}$

graph between log(k) and $\frac{1}{T}$ will be a straight line.

$\log(k) = \log A - \frac{Ea}{2.303R}\left(\frac{1}{T}\right)$

60) Equivalents = Mole × (n–factor) = Faraday
$MnO_4^- \rightarrow Mn^{+2}$; 1 mole × 5 = 5F = 482500 C
$Cr_2O_7^{-2} \rightarrow Cr^{+3}$; 1 mole × 6 = 6F = 579000 C
$Sn^{+4} \rightarrow Sn^{+2}$; 1 mole × 2 = 2F = 193000 C
$Al^{+3} \rightarrow Al$; 1 mole × 3 = 3F = 289500 C

61) **Question Explanation:** Find metal with least tendency to given H_2 from mineral acids.

Concept: Application of electrochemical series.

Solution: Metal with highest value of SRP is least electropositive and has least tendency to give H_2 with mineral acids.
order of SRP → Cu > Ni > Zn > Mn
Therefore Cu has least tendency to give H_2

Final answer: Cu

62) NCERT-XII, Pg. # 94, Part-I, Q. 3.13
From 1st law of electrolysis,
mole × n factor = no. of Faraday

63) NCERT Reference: Class XII, Part-I, Page No. 69

64)

Lead storage battery electrolyte is H_2SO_4 Rusting can be prevented by galvanisation.

65) $w = \frac{E}{96500} \times i \times t$
$E = \frac{63.5}{2}$

66)

$\lambda_m = \lambda_m^o - b\sqrt{C}$
The constant b depends on type of electrolyte. Specifically the charge on cation and anion electrolyte with same charge type (e.g. 1 : 1, 2 : 1, 2 : 2 etc) have the sme value of b.

67)

Oxidising power ∝ SRP value
$KMnO_4$ has highest SRP value so it is powerful oxidising agent.

68)

Explain the Question : We need to calculate Λ^∞ for NH_4OH using the values of Λ^∞ of NH_4Cl, NaOH and NaCl.
Concept : This question is based on Application of Kohlrausch's law.
Solution :
$\Lambda^\infty NH_4OH = \Lambda^\infty NH_4Cl + \Lambda^\infty NaOH - \Lambda^\infty Nacl$
$\Lambda^\infty NH_4OH = 150 + 250 - 130$
$\Lambda^\infty NH_4OH = 270\ ohm^{-1}cm^{-1}\ eq^{-1}$
Final answer : 1

69)

Question Explanation: Determining the direction of conventional current flow in an electrolytic cell's outer circuit.
Concept : This question is based on Conventional current flows opposite to electron flow in the external circuit.
Solution
Electrons flow from anode to cathode. Therefore, conventional current flows from cathode to anode in the external wire (outer circuit).
Final Answer: Option 1

70)

Question Explanation: Find how the pH of $CuSO_4$ solution changes during electrolysis with inert electrodes.
Concept : Electrolysis
Solution:
This question is based on At cathode Cu^{2+} is reduced to Cu, and at anode H_2O oxidizes to O_2 releasing H^+ ions
$Cu^{2+} + 2e^- \rightarrow Cu$ (↓) at cathode
$2H_2O \rightarrow O2 + 4H^+ + 4e^-$ at anode
H^+ increases → pH decreases
Final Answer: Option 2

71)

$H_2 + 2H^+ \rightarrow H_2 + 2H^+$
10^{-3} 10^{-6}

$$Q = \frac{\left(10^{-6}\right)^2}{\left(10^{-3}\right)^2} = 10^{-6}$$

$$E = E° - \frac{0.06}{2} \log (Q)$$

$E = 0 - \frac{0.06}{2} \log(10^{-6}) \approx 0.177V.$

72) **Explain Question:**
What is efficiency of fuel cell.
Concept:
Fuel cell.
Solution:
The efficiency of fuel cell $\eta = \frac{|\Delta G|}{|\Delta H|}$
Final answer: Option (3)

73)

During electrolysis of conc. H_2SO_4, H^+ will be reduced at cathode and SO_4^{-2} will be oxidised at anode.

74) $y_A = \frac{P_A^o x_A}{P}$ and $y_B = \frac{P_B^o x_B}{P}$

$$\frac{y_A}{y_B} = \frac{P_A^o}{P_B^o} \times \frac{x_A}{x_B} \Rightarrow \frac{4}{3} = \frac{1}{3} \times \frac{1-x_B}{x_B}$$

$$x_B = \frac{1}{5}$$

75) $N = \frac{W_B}{M_B \times V} \times (n-\text{factor})$

$$= \frac{1}{98 \times \frac{100}{1000}} \times 2 = \frac{20}{98} = 0.2$$

76)

in air, $P_{N_2} = 0.8\,\text{atm}, P_{O_2} = 0.2$ atm
$P = K_H X$ (Henry's law)
$0.8 = 2 \times 10^4 X_{N_2}$ .....(1)
$0.2 = 10^4 X_{O_2}$ .....(2)
Ratio of solubility of N_2 & O_2 is 2 : 1

77) $\frac{p_0 - p_s}{p_s} = \frac{n}{N}$
$p_s = 75$ & $p° = 100$

78) **Question Explanation:**
Question is asking about choose correct option.
Concept:

This question is based on Henry's law.
Solution:
(1) $P_\gamma = K_H X_\gamma$

$$P_\gamma = 2 \times 10^{-5} \times \frac{55.5}{55.5 + \frac{1000}{18}} = 10^{-5} K \text{ bar}$$

= 0.01 bar (Incorrect)

(2) $P_\delta = K_H X_\delta$

$$P_\delta = 0.5 \times \frac{55.5}{55.5 + \frac{1000}{18}} = 0.25 K \text{ bar}$$

= 250 bar (True)

(3) On increasing temperature solubility of gases decreases (Incorrect)
(4) $K_H \downarrow$ solubility $\uparrow$ and lowest K_H is for γ. (Incorrect)
Final Answer:
The correct option is (2)

79) $\Delta T_f = i.K_f \cdot m$
$0.0054 = i \times 1.8 \times 0.001$
$i = 3$
$\because i = 1 + \alpha (n - 1)$
$3 = 1 + 1 (n - 1)$
$n = 3$

80) $\Delta T_f = \frac{1000}{85}\left(\frac{5}{60} + \frac{10}{180}\right) \times 1.86 = \frac{1000}{85} \times \frac{25}{180} \times 1.86$
$\Delta T_f = 3.04$
$T_f = T_f^o - \Delta T_f$
$= 0 - 3.04 = -3.04°C$

81)

Volume strength of H_2O_2 solution is volume of $O_2(g)$ produced at STP by 1$\because$ of $H_2O_2(aq)$.

82)

$\pi = i \times cRT$
$\pi \propto i \times c$

83) **Explain Question :** We need to determine the vapor pressures of the pure components (A and B).
Concept : This Question is based on Raoult's Law
Solution : From Raoult's Law
$P_{total} = P_A^0 X_A + P_B^0 X_B$ **Case 1:** 1 mole of A ($n_A=1$) and 4 moles of B ($n_B=4$). Total moles = 1+4=5. $X_A=1/5=0.2$ $X_B=4/5=0.8$ $P_{total,1}=560$mm of Hg
Case 2: 1 mole of B is removed. New moles of A ($n_A'=1$) New moles of B ($n_B'=4-1=3$) New total moles = 1+3=4. New $X_A'=1/4=0.25$ New $X_B'=3/4=0.75$ $P_{total,2}=560-10=550$mm of Hg
Equation 1 (Case 1): $560=P_A^0(0.2)+ P_B^0(0.8)$ Multiply by 5: $2800= P_A^0+4 P_B^0$
(i)**Equation 2 (Case 2):** $550= P_A^0(0.25)+ P_B^0(0.75)$ Multiply by 4: $2200= P_A^0+3 P_B^0$ (ii)Now, we have two linear equations:

(i) $P_A^0+4\ P_B^0=2800$
(ii) $P_A^0+3\ P_B^0=2200$Subtract Equation
(ii) from Equation (i): $(P_A^0+4\ P_B^0)-(P_A^0+3\ P_B^0)=2800-2200\ P_B^0=600$mm of Hg
Substitute $P_B^0=600$ into Equation (ii): $P_A^0+3(600)=2200$, $P_A^0+1800=2200 P_A^0=2200-1800$, $P_A^0=400$mm of Hg **Final Answer option (3)**

84)

$i = 1 + (n - 1)\alpha$
$i = 1 + (3 - 1) \times 0.6$
$i = 2.2$
$\Delta T_f = i \times K_f \times m$
$= 2.2 \times 0.2 \times 1.86 = 0.82$
$T_f = T°_f - \Delta T_f = 0 - 0.82$
$T_f = -0.82°C$

85)

$$X_{solute} = \frac{1}{1+55.5} = \frac{1}{56.5}$$
$$X_{solvent} = 1 - \frac{1}{56.6} = \frac{55.5}{56.5}$$
$p_s = p° X_{solvent}$
$$= 92.5 \times \frac{55.5}{56.5} = 90.86 \text{ mm of Hg.}$$

86) **Explain Question:**
Boiling point elevation $\Delta T_b = I \times K_b \times m$. For weak acid, $i = 1 + \alpha$.
Concept:
Relation of van't Hoff factor (i) with degree of dissociation (α).
Solution:
$\Delta T_b = 0.3$, $K_b = 0.5$, $m = 0.4$.
$i = \Delta T_b / (K_b \times m) = 0.3 / (0.5 \times 0.4) = 0.3 / 0.2 = 1.5$
So, $\alpha = i - 1 = 0.5$
Final Answers:
Percentage of dissociation
$= \alpha \times 100 = 0.5 \times 100 = 50\%$

87) $\Delta T_f = iK_fm$
$T_f = T_{f^o} - \Delta T_f$

88)

$CuSO_4 \rightarrow CuSO_{4(aq)}$ $\Delta H = -15.89$
$CuSO_4.5H_2O \rightarrow CuSO_{4(aq)}$ $\Delta H = 2.80$
$CuSO_4 \rightarrow CuSO_4.5H_2O$ $\Delta H = -15.89 - 2.80$

$\Delta H = -18.69$ kCal

89)

Fact Based

90)

Maximum boiling azeotrope are formed by solutions which show negative deviation from ideal behaviour. Water + Nitric acid shows negative deviation.

BIOLOGY

91)

NCERT-XI, Pg. # 58

92) NCERT-XI, Pg. # 64

93) NCERT-XI, Pg. # 59

94)

NCERT-XI, old

95) NCERT-XI, Pg. # 60

96) NCERT-XI, Pg. # 61

97)

NCERT-XI, Pg. # 62

98) NCERT-XI, Pg. # 64

99) NCERT-XI, Old

100) NCERT-XI, Pg. # 67

101) NCERT-XI, Pg. # 64

102) NCERT-XI, Pg. # 67, 68

103)

Old NCERT-XI

104) NCERT XI Pg. # 62

105) NCERT-XII, Pg. # 20

106)

NCERT XI Pg. # 65

107) NCERT XI Pg. # 73

108)

old NCERT XI Pg. # 69

109) NCERT XI Pg # 72

110) NCERT XI Pg # 74

111) NCERT XI Pg # 74

112) NCERT Page No. # 74

113) NCERT XI Pg. # 73, 76

114) NCERT XI Pg. # 76

115) NCERT XI Pg. # 75

116) NCERT XI Page No. # 76,77

117) NCERT-XI, Pg. # 75

118) NCERT XII Pg. # 7-10

119) NCERT XII Page No. # 7,8,22,23

120) NCERT XII, Page No. # 11, 21

121)

NCERT XII, Page No. # 5,9,15,18,19

122) NCERT XII, Page No. # 11-15

123) NCERT XII, Page No. # 11-15

124) NCERT XII, Page No. # 5,8,11,18

125)

NCERT XII Page No. # 5

126) NCERT XII Pg # 13,14

127) NCERT XII Pg. # 12

128)

NCERT XII, Pg. # 11,12

129) NCERT XII Page No. # 11,18

130)

NCERT XII Page No. # 5,9,18

131) NCERT XII Page No. # 8, 9

132) NCERT XII Page No. # 5

133) NCERT XII Page No. # 12

134)

NCERT XII Page No. # 5

135)

NCERT XII Page No. # 5

136) NCERT, Pg # 27

137) NCERT, Pg # 34, 35

138) NCERT, Pg # 28

139) NCERT(XII) Page#34/38H) Para:2.4

140) NCERT Pg. # 49

141) NCERT XII Pg. No. # 37, 38

142)

NCERT Pg. # 49

143) Module

144) NCERT Pg.# 38

145) NCERT P.g No. 30

146) Conduction of gametes is function of secondary sex organs.

147) NCERT(XII) Chapter #2

148) NCERT(XII) Chapter # 2

149) NCERT Pg. # 53

150) NCERT(XII) Page#37/41(H) Para:2.6

151) Module

152) NCERT(XII) Page#35/38(H) Fig:2.5

153) NCERT(XII) Page#36/39(H) Para:2.5

154) NCERT Pg # 30

155) NCERT XII Pg. No. # 33

156) NCERT (XII) ; Page No. # 31

157) NCERT Pg. # 35

158) NCERT PAGE (ENGLISH): 28,30

159) NCERT(XII) Page#32/35(H) Para: 2.3

160) NCERT(XII) Page#33/36(H) Fig:2.8

161) NCERT(XII) Page#28/30(H) Para:2.1

162) NCERT(XII) Page#28/30(H) Para:2.1

163) NCERT Pg. # 31

164) NCERT-XII, Pg. # 46

165)

NCERT Pg. # 60

166) NCERT(XII) Page#45/50(H) Para:3.2

167) NCERT(XII) Page#48/54(H) Para:3.5

168) NCERT(XII) Page#45/51(H) Para:3.2

169) NCERT(XII) Page#44/50(H) Para:3.2

170) NCERT P.g No. 48

171) NCERT-XII, Pg # 48

172) NCERT Page No.: 44

173) NCERT Pg. # 47

174) NCERT Pg # 46

175) NCERT Pg # 48, 49

176) NCERT Pg # 45-46

177) NCERT, Pg # E-41, H-46, 47

178) NCERT Pg. # 47

179) NCERT Pg. # 45

180) Copper IUDs release copper ions that impair sperm motility and fertilization.